
Machine Learning for Plasma Reactor Optimisation: Predicting and Maximising Ozone Production

Graduation Internship Report

Bachelor of Science - Data Science & Artificial Intelligence

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Abstract

KapiteinLabs is a startup that develops plasma reactors for ozone production, where short high-voltage pulses are applied to a flow of air to generate ozone for water and air treatment applications. Designing a reactor for a specific ozone output is currently a slow process that depends on running new experiments for every configuration. This project develops a machine learning pipeline that supports reactor design and testing in two ways: a prediction tool that estimates ozone output from the controllable reactor parameters, and an inverse optimisation tool that recommends reactor settings for a user-defined target ozone output.

The work is based on the Campaign 5 dataset, a series of pulsed-power plasma reactor experiments conducted at TU Eindhoven covering 438 measurements across six experimental sessions after data cleaning. Five models were trained and compared on the same train/test split: Linear Regression, Gradient Boosting, XGBoost, a Neural Network, and a Physics-Informed Neural Network (PINN) with monotonicity constraints derived from the plasma ozone literature. The PINN achieved the best test set performance, with $R^2 = 0.988$, $\text{MAE} = 1.48 \text{ g/kWh}$, and $\text{MAPE} = 1.9\%$. Permutation importance analysis showed that reactor length and repetition rate together account for around two-thirds of the predictive signal. The inverse optimisation framework, built on top of the trained models using differential evolution, recovers reactor settings that hit user-defined yield targets to within 0.01 g/kWh in most cases.

The full pipeline, including the trained models, the inverse optimisation tool, and a Streamlit dashboard for non-technical users, was delivered to KapiteinLabs. The dashboard supports retraining on new experimental data and is designed to grow as the dataset expands.

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1 — Introduction

1.1 Background and Context

Ozone is a gas used to clean water and air. It works by breaking down harmful substances such as bacteria, viruses, and chemical pollutants. Because of this, ozone is widely used in drinking water treatment, in industrial cleaning processes, and in air purification systems. One important property of ozone is that, because it is chemically unstable and decomposes back into oxygen with a short half-life, under most conditions it is difficult to store or transport (Batakliiev et al., 2014). It therefore usually has to be produced on the spot, at the place where it will be used. This means every ozone application needs its own ozone generator, and the size and efficiency of that generator directly affects the cost and performance of the system.

KapiteinLabs is a startup that develops plasma reactors for ozone production. The reactors work by sending short high-voltage electrical pulses through a flow of air inside a metal tube. These pulses create a plasma, which is a partially ionized state of the gas in which some of the oxygen molecules are split apart. The split oxygen atoms then combine with other oxygen molecules to form ozone. The amount of ozone produced, and how much electricity it takes to produce it, depends on several settings: the length and diameter of the tube, the voltage of the pulses, how many pulses per second are sent, the width of each pulse, and how fast the air flows through the tube. Finding the best combination of these settings is currently done by running many experiments, which takes time and is expensive.

This graduation project was carried out at KapiteinLabs during a six-month internship. The goal of the project is to use machine learning to support the work of finding good reactor settings. The project has two main parts. The first part is a prediction tool: given a set of reactor settings, the tool predicts the reactor's ozone performance, such as its ozone yield and concentration. This allows engineers to test different settings on the model first and only run the experiments that look promising, instead of running every combination. The second part is an optimisation tool that works the other way around: the user enters a target amount of ozone, and the tool returns the reactor settings that should produce that amount. This is useful for both KapiteinLabs and its clients. Internally, it helps the team plan experiments. Externally, it lets clients estimate what kind of reactor they would need for their specific ozone demand, without having to run a full set of experiments first.

1.2 Problem Statement

Designing a plasma reactor for a specific ozone application is currently a slow and experiment-heavy process. Each new reactor configuration, defined by the tube length, diameter, and pulse settings, has to be tested in the lab to find out how much ozone it produces and how efficient it is. Running these experiments takes time, requires specialised equipment, and is expensive.

For KapiteinLabs the situation is more difficult than for a larger research group, because the company does not run most of the experiments itself. The data used in this project, for example, comes from experiments carried out at TU Eindhoven. This means KapiteinLabs depends on external partners for new measurements and datasets, which limits how quickly the company can answer specific questions about reactor performance, especially for configurations that have not been tested before.

This creates two practical problems. The first is internal: when planning new experiments, there is no fast way to estimate which parameter ranges are worth testing. As a result, time and resources are spent on combinations that turn out to be uninformative. The second is commercial: when a potential client asks how large a reactor they would need to produce a given amount of ozone, there is currently no quick way to give a reliable answer. Each estimate either has to rely on past experiments with similar settings or wait for new ones to be run.

A machine learning model that predicts ozone output from reactor settings, and an optimisation tool that works the other way around, would address both problems. It would allow KapiteinLabs to plan experiments more selectively and to give clients faster, data-driven estimates of the reactor configuration that fits their needs. This project develops and evaluates such a model and tool.

The challenge of optimising plasma reactor design is not unique to KapiteinLabs. Plasma discharge technology has been studied for many years as a way of producing ozone and other reactive species for water treatment, and the relationship between operating parameters and reactor performance is non-linear and depends on many interacting factors such as reactor geometry, electrode configuration, energy input, and gas composition (Jiang et al., 2014). More recent work has started to apply machine learning to plasma reactor design directly. Shi et al. (2024) showed that combining a data-driven model with an optimisation search can be both feasible and useful for thermal plasma reactor geometry, and Li et al. (2025) used a machine learning model to both predict and optimise the performance of a dielectric barrier discharge reactor for methane reforming, based on experimental data. Both studies follow the same two-part pattern used in this project, a data-driven predictor combined with an optimisation step, and both confirm that machine learning can capture the non-linear behaviour of plasma reactors well enough to guide their design. What these studies have in

common is that none of them target pulsed-plasma ozone reactors operating on air at atmospheric pressure. The gap this project addresses is therefore the absence of a predict-and-optimise machine learning workflow for this specific reactor type, where the controllable parameters and the underlying physics differ from both the thermal plasma geometry of [Shi et al. \(2024\)](#) and the catalytic methane process of [Li et al. \(2025\)](#). This project addresses that gap with a workflow built around the wire-cylinder pulsed-plasma reactor.

1.3 Scope and Limitations

This project covers the development of a machine learning pipeline for predicting ozone output from plasma reactor settings, an optimisation tool that recommends reactor settings for a target ozone output, and a dashboard that makes both tools available to users at KapiteinLabs. The dashboard is delivered with the trained models included and also allows the user to upload new experimental data and retrain the models on it, so that the tool can stay useful as more measurements and datasets become available.

The work is based on data from experiments carried out at TU Eindhoven on wire-cylinder configuration reactors using synthetic air. The dataset covers a limited range of reactor sizes and operating conditions, which are described in detail in Chapter 3. Predictions are only reliable inside this range. The tool can return an answer for settings outside it, but those answers should be treated as rough estimates rather than trustworthy predictions. Three limitations of the project should be highlighted.

The first is the size and coverage of the dataset. After cleaning, the dataset contains 438 measurements from six experimental sessions. This is enough to train accurate models for the configurations that were tested, but it is not enough to cover the full range of reactor designs that KapiteinLabs may want to build in the future. In particular, only two reactor diameters and two flow rates were tested, which makes it hard for the model to predict performance for new geometries or operating conditions outside this range.

The second is the dependence on external experiments. KapiteinLabs does not run most of the experiments itself, so adding new data to the dataset depends on the schedule of external research partners. This limits how quickly the model can be improved or extended to new reactor types.

The third is the limited number of sensors used during the experiments. Only electrical signals and ozone concentration were recorded. Other variables that are known to affect plasma behaviour, such as gas temperature and humidity, were not measured. Adding these sensors in future experiments would allow the model to capture more of the physical process and make predictions more robust to changes in environmental conditions.

Economic factors such as the cost of electricity, reactor manufacturing, or maintenance are also out of scope. The tool predicts ozone output and recommends settings to reach a target output, but it does not evaluate whether a given configuration is commercially attractive.

1.4 Research Questions

Building on the literature gap identified in the problem statement, this section formulates the research questions that guide the rest of the report.

The main research question of this project is:

How well can machine learning models trained on plasma reactor experiment data predict ozone output and recommend reactor settings for a target ozone production?

This main question is supported by three sub-questions:

1. How accurately can a machine learning model predict ozone yield from the controllable reactor parameters, and which model type performs best on this dataset?
2. What impact does adding physical knowledge to the model in the form of monotonicity constraints have on prediction accuracy and physical consistency?
3. How well can an inverse optimisation method, built on top of the predictive model, recommend reactor settings that produce a user-defined target ozone output within the trained parameter range?

1.5 Report Structure

Chapter 2 introduces the plasma reactor physics, ozone chemistry, and the key feature relationships needed to interpret the dataset. Chapter 3 describes the dataset, the data cleaning process, the machine learning models developed for the prediction task, and the inverse optimisation framework built on top of them. Chapter 4 reports and discusses the experimental results. Chapter 5 describes the interactive dashboard built on top of the model. Chapter 6 draws conclusions and provides recommendations.

2 — Literature Review

This chapter covers the scientific background relevant to the project: how plasma reactors produce ozone, what governs ozone yield, and how the experimental parameters connect to the machine learning features.

2.1 How Ozone is Produced in a Plasma Reactor

The plasma reactors used in this project rely on pulsed power to generate ozone. In a pulsed system, very short high-voltage pulses (on the order of nanoseconds) are applied to the reactor rather than a continuous voltage. This approach has several advantages over non-pulsed plasma generation. Shorter pulses produce higher ozone yields because they concentrate the electron energy into brief, highly reactive events that efficiently dissociate oxygen molecules. Pulsed operation also makes it possible to develop and stabilise non-equilibrium plasmas at atmospheric pressure: without pulsing, the plasma temperature would exceed 1000 K and the discharge would become unstable at atmospheric conditions. In addition, the short pulses generate highly reactive radicals that are effective not only for ozone production but also for air purification applications more broadly (Kebriaei et al., 2015).

Figure 2.1 illustrates this principle.

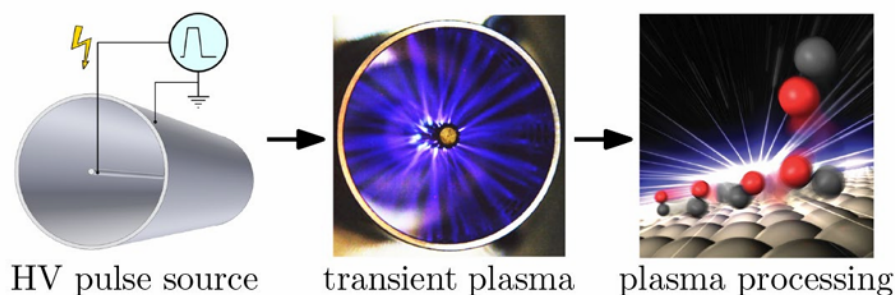


Figure 2.1: Pulsed plasma ozone generation. A high-voltage pulse source (left) delivers nanosecond pulses to a wire-cylinder reactor, igniting a transient plasma (centre) that produces reactive species for ozone formation (right). Figure courtesy of TU Eindhoven, originally presented by Ton (2021).

Specifically, the reactor used in the experiments is a wire-cylinder configuration: a 0.2 mm thick high-voltage steel wire runs along the centre of a cylindrical metal tube. When a pulse is applied, the strong electric field near the wire accelerates free electrons to high energies. These energetic electrons collide with oxygen molecules, dissociating them into reactive atomic oxygen:



The resulting atomic oxygen then reacts with a nearby oxygen molecule to form ozone (Huiskamp et al., 2017):



where M is any third molecule that absorbs the excess energy.

Because the experiments use synthetic air rather than pure oxygen (synthetic air is the standard feed gas for industrial ozone generators, since using pure oxygen would be significantly more expensive and logistically impractical at scale), a fraction of the electron energy is consumed by nitrogen reactions rather than ozone-forming ones. At high energy densities this also leads to NO_x formation, which reacts with and destroys ozone, an effect known as *discharge poisoning* (Huiskamp et al., 2017).

This is the physical reason why ozone yield decreases at high energy density, which is one of the key patterns in the dataset.

2.1.1 Air vs. Pure Oxygen

All experiments in this project used dry synthetic air (approximately 21% O_2 , 79% N_2) rather than pure oxygen. In pure oxygen, all electron energy can go toward ozone formation. In air, a significant fraction is consumed by nitrogen reactions, and some of the resulting nitrogen species actively destroy ozone through discharge poisoning (Huiskamp et al., 2017). The practical consequence is that ozone yields (i.e. grams of ozone per kWh) in air are lower than in pure oxygen at the same energy density, and the yield decrease at high energy densities is more pronounced. Since all data in this project come from air experiments, the model learns the air-specific yield behaviour and predictions are only valid for air operation.

2.1.2 Streamer Phases and Why They Matter

The high-voltage pulse does not simply create a uniform plasma. Instead, it triggers *streamers*: fast-moving ionisation fronts that propagate from the wire outward toward the grounded cylinder wall. Streamer propagation happens in two distinct phases (Ton, 2021):

The **primary phase** is fast. Streamers cross the reactor at high velocity and produce the most energetic electrons. This is the most efficient phase for ozone production because the electrons are fast enough to trigger reaction 2.1 efficiently.

The **secondary phase** is slower. It occurs after the primary streamers have crossed the gap and involves a residual lower-energy discharge near the wire. It draws significant current but produces far fewer oxygen radicals per joule because the electrons are less energetic.

The practical consequence is that whenever the secondary phase dominates, more energy is consumed for less ozone. Ton (2021) showed that operating at a voltage just barely above the streamer ignition threshold prevents the secondary phase from occurring entirely, producing very high ozone yields at low energy density. However, this optimal threshold depends on the specific reactor geometry (tube length, diameter) and operating conditions (flow rate, repetition rate, pulse width and energy density), making it difficult to determine analytically for new configurations. This is precisely the gap that the machine learning models developed in this project aim to fill: rather than requiring extensive manual experimentation for each new reactor setup, a trained model can predict the expected ozone yield across the full parameter space and identify the optimal operating point automatically. This non-linear, non-monotonic relationship between voltage and ozone yield is one of the most important physical patterns in the dataset and is a key reason why linear models fail to capture it well.

2.2 Ozone Yield vs. Energy Density: The Key Relationship

The central relationship in the dataset is between ozone yield [g/kWh] and energy density [J/L]. Energy density describes how much energy each litre of gas receives from the plasma, calculated as:

$$\varepsilon = \frac{f_{rr} \cdot E_p \cdot 60}{F} \quad (2.3)$$

where f_{rr} is the repetition rate [Hz], E_p is the energy per pulse [J], and F is the gas flow rate [slm]. Ozone yield is then defined as:

$$G_{O_3} = \frac{C_{O_3} \cdot FW_{O_3} \cdot 3.6}{V_m \cdot \varepsilon} \quad (2.4)$$

where C_{O_3} is the ozone concentration [ppm], $FW_{O_3} = 48$ g/mol is the formula weight of ozone, and $V_m = 24.5$ L/mol is the molar volume at room temperature (Huiskamp et al., 2017; Ton, 2021).

Figure 2.2 shows this relationship from a single experimental session (a session refers to one continuous set of measurements using a fixed reactor configuration, in this case a 1 m long, 50 mm diameter reactor at 20 L/min flow rate). Yield is highest at low energy densities and decreases as energy density increases. Each colour represents a different repetition rate and each symbol a different voltage setting. The plot shows that the same energy density can produce different yields depending on how it was achieved, which reflects the interplay between voltage, repetition rate, and streamer phase described above.

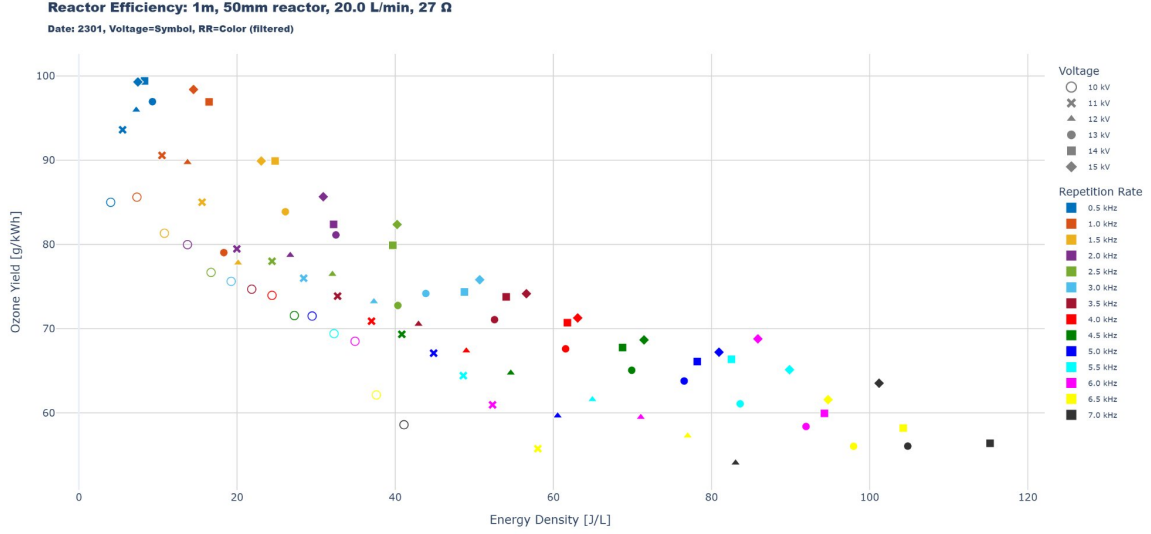


Figure 2.2: Ozone yield [g/kWh] vs. energy density [J/L] for a single session (reactor: 1 m, 50 mm, 20 L/min). Each symbol represents a voltage level (10–15 kV) and each colour a repetition rate (0.5–7 kHz). Yield decreases with energy density due to discharge poisoning and increased secondary-streamer activity at higher voltages and repetition rates.

An important consideration for modelling is that energy density is calculated from input-side electrical measurements. Because of impedance mismatch between the pulse source and the reactor (the reactor behaves as a transmission line at nanosecond timescales), some of the input energy reflects back rather than coupling into the plasma (Huiskamp et al., 2016). The energy density feature therefore slightly overestimates the energy actually deposited, and the degree of overestimation depends on the reactor geometry. This is why geometric features (reactor length, diameter) need to be included as model inputs alongside energy density rather than relying on energy density alone.

2.3 Features: What They Represent and Why They Are Used

The six controllable input features in the dataset map directly to the physical parameters that determine how the plasma operates.

Voltage setting [kV] controls the electric field strength near the wire. A higher voltage produces more energetic electrons and a larger streamer volume, which generally increases ozone production up to the point where the secondary streamer phase begins. Above that threshold, further increasing voltage reduces efficiency because more energy goes into the inefficient secondary phase (Ton, 2021).

Repetition rate [kHz] controls how many pulses per second are delivered to the reactor. Higher repetition rate increases the energy density (via equation 2.3) without changing the plasma physics within each individual pulse. This makes it a more efficient way to scale up throughput than raising voltage, because it does not push the operating point toward the secondary streamer phase (Ton, 2021).

Flow rate [L/min] controls how long the gas spends inside the reactor. A lower flow rate means each litre of gas is exposed to more pulses before leaving the tube, increasing its energy density. Experimental results from Ton (2021) showed that flow rate does not significantly affect ozone yield at the same energy density, which means it primarily acts as a throughput scaler rather than an efficiency driver.

Reactor length [m] determines the active plasma volume and is related to how many streamers form in parallel along the tube, the energy dissipated per pulse, and the impedance seen by the pulse source (Huiskamp et al., 2016; Ton, 2021). These effects pull ozone yield in opposite directions, so the net effect on yield depends on the specific reactor geometry and operating settings.

Reactor diameter [mm] affects the ignition voltage threshold. A wider tube requires a stronger field to initiate streamers because they must travel further to cross the gap. This shifts the optimal operating voltage upward and changes the ozone yield behaviour as a function of energy density (Ton, 2021).

Pulse width (FWHM) [μ s] describes the duration of the active plasma discharge, measured as the full width at half maximum of the voltage pulse. A shorter pulse concentrates the electron energy into a briefer, more intense event, which favours the efficient primary streamer phase over the less productive secondary phase. Pulse width varies with the pulse source settings and interacts with reactor geometry, since longer or wider tubes change the impedance seen by the pulse source and therefore affect the pulse shape delivered to the plasma. Because it affects the pulse shape, pulse width also affects the energy per pulse, and therefore the energy density delivered to the gas (equation 2.3).

2.4 Measurement Approach and Data Origin

Ozone concentration was measured by UV-absorption spectrometry using the Beer-Lambert law applied to the 200–260 nm Hartley absorption band of ozone (Ton, 2021), which is the wavelength range where ozone absorbs UV light most strongly. The Beer-Lambert law states that light absorption is proportional to the concentration of the absorbing species and the optical path length:

$$A = -\log\left(\frac{I}{I_0}\right) = \sigma \cdot N \cdot L \quad (2.5)$$

where A is absorbance, I/I_0 is the ratio of transmitted to incident light intensity, σ is the absorption cross-section of ozone [cm^2], N is the ozone number density [cm^{-3}], and L is the optical path length [m]. The ozone concentration in ppm is then derived from N using the ideal gas law at room temperature and atmospheric pressure.

Electrical features were extracted from oscilloscope waveforms recorded during each pulse. The voltage waveform was reconstructed from the D-dot probe signal (a capacitive dV/dt measurement) by applying an inverse transfer function that undoes the hardware integrator response. Energy per pulse was then calculated by numerically integrating the product of voltage and current over the pulse duration.

The entire analysis pipeline was originally written in MATLAB by the TU Eindhoven research group. For this project, the pipeline was fully reimplemented in Python to enable direct integration with the machine learning workflow. This reimplementation involved translating the signal processing logic used to correct raw voltage measurements (including applying an inverse transfer function with specific filter coefficients), recreating the numerical integration routines for calculating energy per pulse, and rebuilding the UV spectrometry analysis that extracts ozone concentration from raw spectral data at 17 wavelengths. The Python pipeline also had to handle the file-matching logic between electrical and spectrometry data across all 12 experimental dates, accounting for per-date exceptions in measurement numbering and folder structure. In total, the reimplementation covered approximately 1,500 lines of processing code spread across multiple notebooks. Results were validated against the original MATLAB outputs before proceeding to model training.

3 — Methodology

3.1 Dataset Description

The dataset used in this project comes from a series of plasma reactor experiments referred to as Campaign 5, conducted at TU Eindhoven by Joris Bonnema. The term “campaign” is used by the research group to denote a complete round of experiments with a shared experimental design; Campaign 5 is the fifth such round and the most recent one available at the time of this project.

Figure 3.1 shows the experimental setup at TU Eindhoven used to collect the Campaign 5 data.

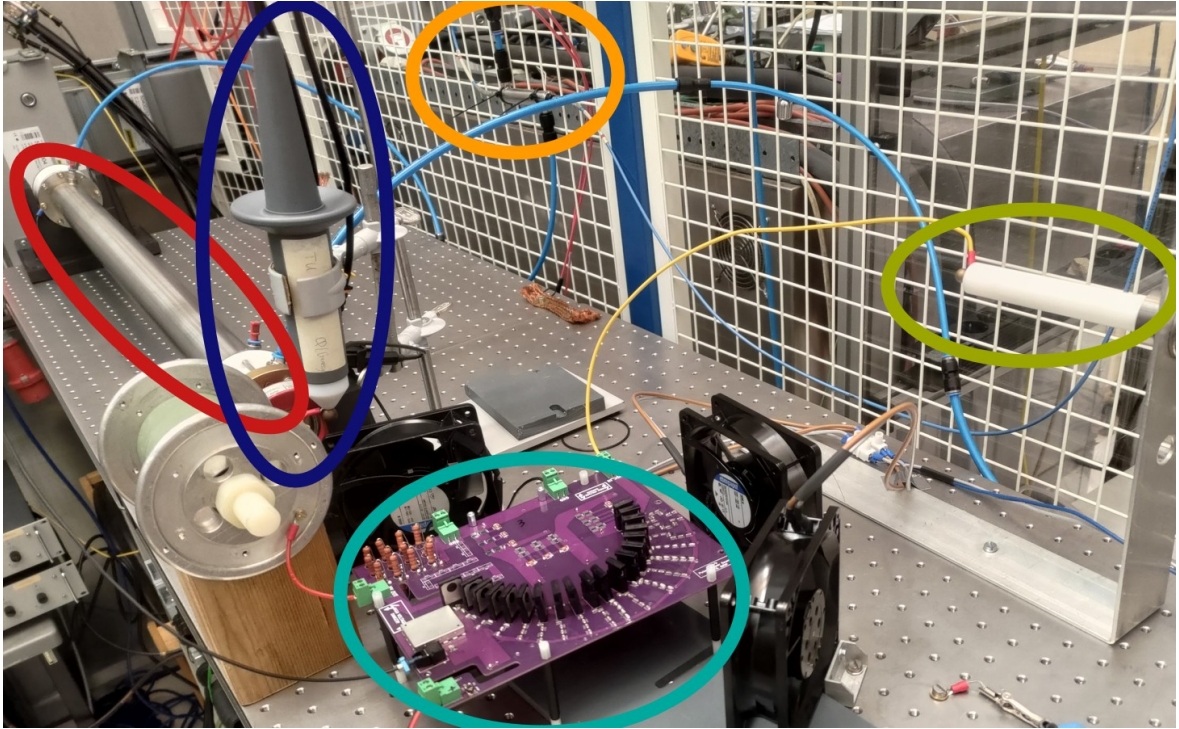


Figure 3.1: Experimental setup at TU Eindhoven for the Campaign 5 ozone measurements. Coloured circles indicate the main components: plasma reactor (red), HV switch (teal), spectrometer (orange), HV supply (olive), and sensors (blue). Photo courtesy of Joris Bonnema.

The experiments were carried out across 12 experimental sessions spanning multiple dates, with each session using a specific reactor configuration and sweeping over combinations of voltage (10–15 kV), repetition rate (0.5–7 kHz), and pulse width (0.09–0.2 μ s). Reactor configurations varied between sessions in terms of tube length (12.5 cm to 2 m) and inner diameter (30 mm to 50 mm), as well as air flow rate (10 or 20 L/min).

The raw data consists of two entirely separate sources that had to be matched and combined for every measurement. Figure 3.2 shows the overall folder layout of the raw data.

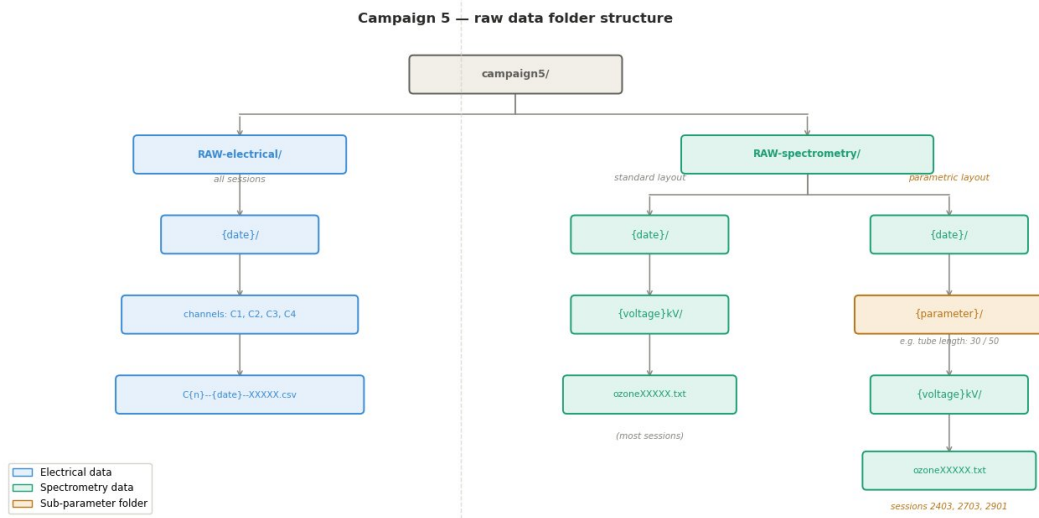


Figure 3.2: Raw data folder structure. Electrical data (blue) is stored per date with one CSV file per oscilloscope channel per measurement. Spectrometry data (green) is organised by date and voltage level. Three sessions used a parametric sub-folder layout with an additional reactor parameter level before the voltage folders.

Electrical data was recorded by an oscilloscope at 4 channels per measurement. Channel 2 captured the D-dot probe output (a dV/dt signal), channel 3 captured the current transformer output (1 V/A), and channels 1 and 4 captured a PMK probe reference and the trigger signal respectively. Each measurement was stored as a separate CSV file per channel, named by date and sequential index (e.g. `C2--2301--00042.csv`). Across all 12 dates the electrical data comprised over 2000 individual oscilloscope files.

Spectrometry data was recorded in parallel using UV-absorption spectroscopy. Each measurement produced a `.txt` file containing the UV absorption spectrum of the gas sampled from the reactor outlet, with wavelength on the x-axis and light intensity on the y-axis. The ozone concentration in ppm was derived from the absorption spectrum using the Beer-Lambert law at 17 wavelengths in the 200–260 nm Hartley band, using known ozone absorption coefficients. The spectrometry files were organised into subfolders by voltage level within each date folder.

Matching electrical and spectrometry measurements was done by file index: a spectrometry file `ozone00042.txt` corresponds to electrical files `C1--2301--00042.csv` through `C4--2301--00042.csv`. This index-based pairing was the only available alignment method, as no explicit metadata file linked the two sources.

Figure 3.3 shows the voltage coverage per session and the total file count per session for both data types. The large variation in file count across sessions reflects differences in the number of voltage levels tested and the number of repetition rate steps per level. One session stands out with nearly 1800 spectrometry files, as it covered two reactor configurations (30 mm and 50 mm diameter) within the same date. Three of the twelve sessions used a different folder layout (a parametric sub-folder structure with an additional reactor parameter level before the voltage folders). This was relevant for the data processing pipeline because it required separate file-matching logic for those sessions; without handling this exception, those measurements would have been incorrectly paired or skipped entirely.

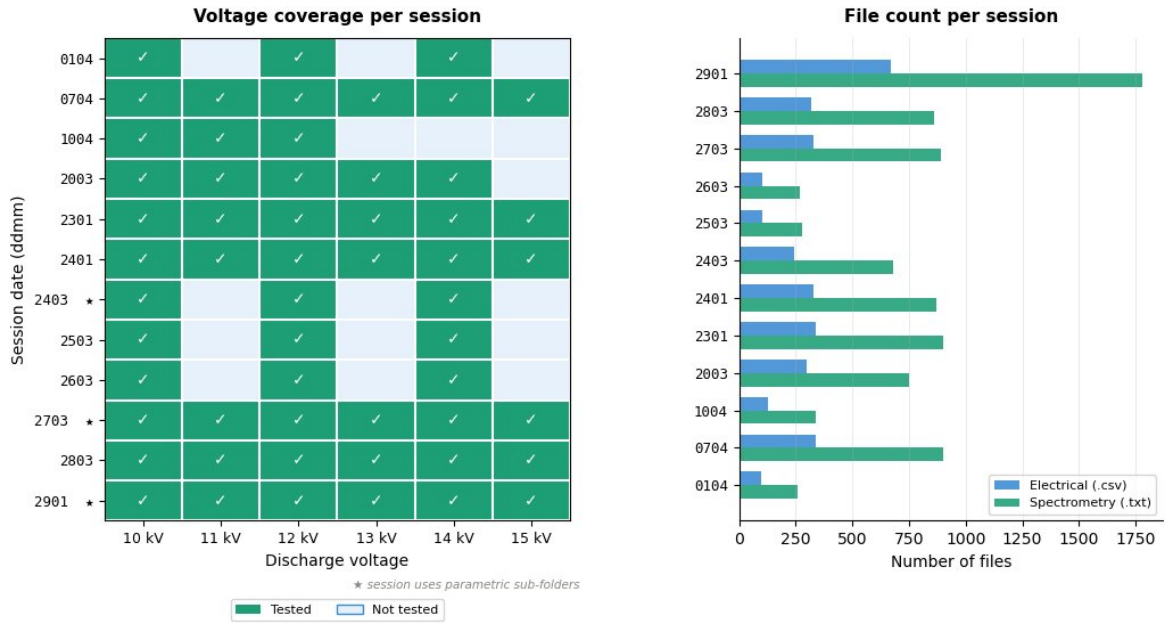


Figure 3.3: Left: voltage coverage per experimental session. Green cells indicate voltage levels that were tested. Right: total file count per session for electrical (.csv) and spectrometry (.txt) files.

Data origin and responsible use. The Campaign 5 dataset was collected by the TU Eindhoven research group and shared with KapiteinLabs for the purpose of supporting reactor design work. The data was used in this project with the awareness of the original experimenters at TU Eindhoven, and design decisions that affect the dataset (such as the exclusion of the 12.5 cm reactor sessions discussed in Section 3.4) were made in direct consultation with the original experimenter, Joris Bonnema. The dataset contains no personal data and the experiments do not involve human or animal subjects, so the main ethical considerations for this work are honest reporting of the preprocessing decisions and clear attribution of the data source. Both are documented in the following sections, where every exclusion is recorded with its reason and every step of the analysis pipeline is described in enough detail to be reproduced.

3.2 Data Preprocessing Challenges

The raw data was one of the main practical challenges of this project. The raw files were highly unstructured, with no standardised documentation, inconsistent folder naming, varying numbers of measurements per voltage level, and per-date exceptions in the sweep order and measurement count. The only available reference was a set of experimental notes compiled by the original experimenter, which documented reactor settings and flagged known anomalies such as missing data points and non-standard sweep orders on specific dates.

At the start of the project it was unclear whether the electrical and spectrometry files could be reliably matched by index, since the two acquisition systems operated independently. This was identified as a project risk: if the index alignment failed, the only fallback would have been to manually extract data points from the plots produced by the original MATLAB analysis, which would have been far less accurate and significantly more labour-intensive. The risk was resolved by cross-validating the matched dataset against the original MATLAB output plots, confirming that the index-based pairing produced results consistent with the prior analysis. Figure 3.4 shows an example of this validation for one session (100 cm, 50 mm reactor, 20 L/min): the Python-generated ozone yield vs. energy density plot (bottom) closely reproduces the MATLAB reference plot (top), confirming that the pipeline reconstruction was successful.

The original analysis had been performed in MATLAB using scripts written by the TU Eindhoven researchers. These scripts were studied to understand the data structure and processing logic, and the entire pipeline was then recreated in Python to enable flexible manipulation and machine learning integration. Comparison plots between the Python outputs and the original MATLAB figures were used to verify that the reimplementation was correct before the dataset was used for modelling.

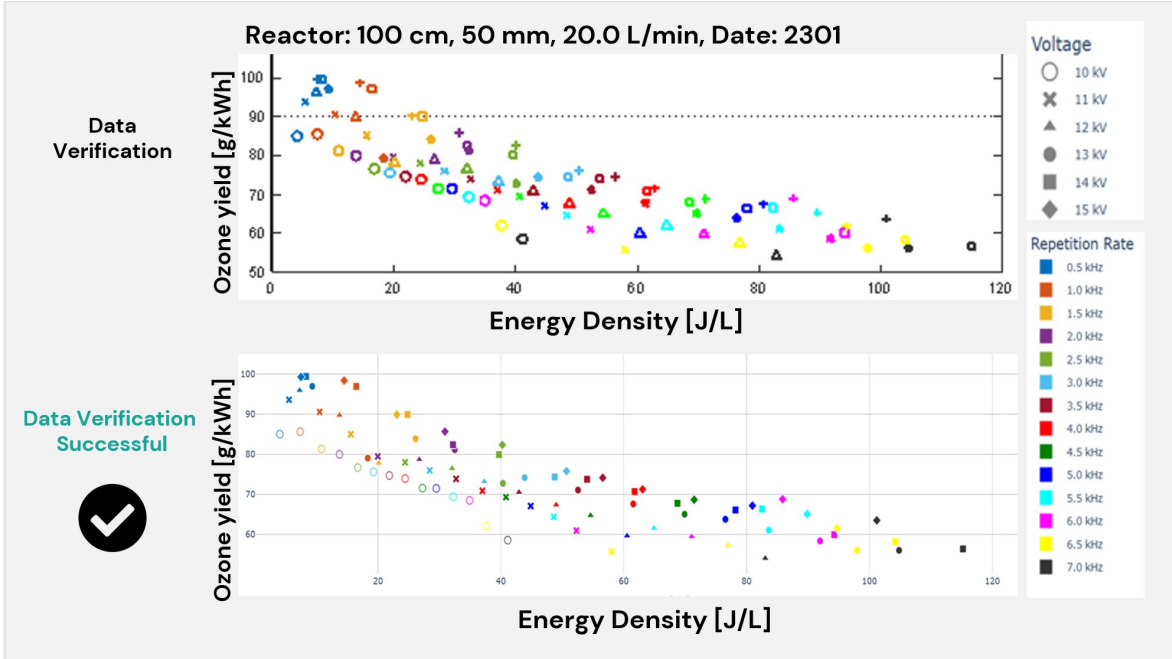


Figure 3.4: Data verification for one session (reactor: 100 cm, 50 mm, 20 L/min). Top: original MATLAB output. Bottom: Python reimplementation output. The close match in pattern and value range confirms that the Python pipeline correctly reproduces the original analysis.

3.3 Feature Engineering and Target Variable

Each matched electrical–spectrometry pair was processed as follows. The raw D-dot probe signal (channel 2) was corrected using a transfer function that inverts the hardware integrator response to recover the actual voltage waveform. The energy per pulse E_p was then calculated by numerically integrating the product of voltage $V(t)$ and current $I(t)$ over the full pulse window, from the start of the recorded signal to the end. This window is wider than the FWHM pulse width `fwhm_us`, which is reported as a separate feature and only measures the half-maximum width of the voltage pulse. From E_p , the energy density ε [J/L] was derived as $\varepsilon = (f_{rr} \cdot E_p \cdot 60) / F$, where f_{rr} is the repetition rate [Hz] and F is the flow rate [slm]. The ozone concentration in ppm was extracted from the spectrometry files using the Beer-Lambert calculation described by [Ton \(2021\)](#), and converted to ozone production rate in g/h using the flow rate and molar mass of ozone.

The full feature set of the dataset can be divided into four groups based on their origin and role.

Identifiers (`date_id`, `description`) label each row with the experimental session it belongs to and are not used as model inputs.

Controllable parameters are the six variables that can be directly set before an experiment and reproduced in future experiments:

- `reactor_length_m` – active tube length (0.125–2.0 m)
- `reactor_diameter_mm` – inner tube diameter (30–50 mm)
- `voltage_setting_kV` – supply voltage set on the pulse source (10–15 kV)
- `repetition_rate_kHz` – pulse repetition rate (0.5–7.0 kHz)
- `flow_rate_L_per_min` – dry air flow rate through the reactor (10 or 20 L/min)
- `fwhm_us` – full width at half maximum of the voltage pulse (0.09–0.2 μ s)

These are the only features that can be used as actionable inputs in the inverse optimisation framework, since they are the only ones a reactor operator can actually set. This is an important limitation: although the dataset contains many additional electrical features measured by the oscilloscope, those features are consequences of the pulse dynamics and cannot be independently prescribed.

Electrical features are derived from the oscilloscope waveforms (channels C2 and C3) during each pulse. They include peak voltage, minimum voltage, peak-to-peak voltage swing, peak and mean current, peak instantaneous power, mean power, maximum voltage rise rate, energy per pulse, dissipated energy in the shunt resistor,

and peak capacitive current. These features are outputs of the interaction between the pulse source settings and the plasma reactor load and cannot be directly controlled or prescribed.

Derived features are calculated from the above using simple arithmetic. They include energy density [J/L] (energy per pulse times repetition rate divided by flow rate), average continuous power [W], reactor volume [m³], and gas residence time [s].

The dataset contains four target variables, each describing ozone production from a different perspective:

- **ozone_ppm** [ppm] – ozone concentration in the output gas, measured directly by UV spectrometry via the Beer-Lambert law. This is the primary target and the closest to the raw measurement, with the least accumulated calculation error.
- **ozone_yield_g_per_kWh** [g/kWh] – energy efficiency: grams of ozone produced per kilowatt-hour of electricity consumed. The key metric for industrial applications where energy cost matters. Yield typically decreases as energy density increases, reflecting the trade-off between throughput and efficiency discussed in the literature review.
- **ozone_concentration_g_per_m3** [g/m³] – ozone mass concentration, a simple unit conversion from ppm ($\times 47.997 / 24.5$). Useful for applications requiring a specific mass per volume, such as water treatment.
- **ozone_production_rate_g_per_h** [g/h] – total ozone output per hour, combining concentration with volumetric flow rate. This is the throughput target and the most relevant specification for industrial reactor sizing.

3.4 Data Cleaning

The raw Campaign 5 data comprised 12 experimental sessions totalling approximately 726 individual measurements. Reaching the final training dataset required four successive cleaning stages, summarised in Table 3.1.

Stage 1: Exclusion of non-comparable sessions

Four of the twelve sessions were excluded from the dataset entirely. Session 0104 used a different circuit configuration (lower inductance) that changed the pulse shape delivered to the reactor, making its electrical features incomparable with the remaining sessions. Sessions 2503, 2403 (check 1), and 2403 (check 2) were repeatability checks performed under modified conditions (different UV source, compressed air instead of synthetic air) and were not included in the main dataset. Together these four sessions accounted for 95 measurements.

Stage 2: Incomplete sweeps and individual measurement exclusions

Within the eight remaining sessions, not every voltage/repetition-rate combination was measured. Several voltage levels were run as *short* sweeps in which the highest one to three repetition rates were omitted because the experiment was stopped early. For example, session 2803 at 15 kV only covered repetition rates up to 5.5 kHz instead of the full 7.0 kHz. In session 1004 (2 m reactor) only three voltage levels were tested and the 12 kV sweep was limited to three repetition rates. In session 2301, electrical data file #41 was missing from the raw data folder and was therefore skipped during loading.

Two individual ozone measurements were also excluded based on flags in the original experimental notes. In session 2703 (25 cm, 40 mm reactor) the spectrometry files for measurement #50 at 13 kV contained anomalous readings and were removed. In session 2803 (1 m, 40 mm reactor) the first ozone measurement at 13 kV was flagged as unreliable and discarded. In total, Stage 2 reduced the eight included sessions from 631 to 604 rows.

Stage 3: Exclusion of the 12.5 cm reactor

The 12.5 cm reactor sessions (date 2901, covering both the 50 mm and 30 mm diameter tubes) were excluded from the final training dataset. This decision was made after consultation with Joris Bonnema (TU Eindhoven), who indicated that measurements from the shortest reactor tube are less reliable than those from longer tubes due to the limited active plasma volume and stronger edge effects at that scale. The effect

of this exclusion on model performance is evaluated in Section 3.6.7. Excluding these two sub-sessions removed 166 rows, bringing the dataset from 604 to 438 rows.

Stage 4: Post-processing filters

Four automated filters were applied to the combined dataset to remove any remaining physically invalid observations:

1. Rows with missing ozone values (failed spectrometry reads).
2. Rows with negative ozone concentration, which is physically impossible and results from spectrometer noise in the baseline measurement.
3. Rows with very low energy per pulse ($E_p < 0.02$ mJ), corresponding to pulses where the plasma did not properly ignite.
4. Rows with ozone yield outside the range 0–500 g/kWh, which are not physically meaningful and arise from dividing by a near-zero energy density.

In the current dataset build, all four filters removed zero additional rows, confirming that the exclusions in Stages 1–3 had already eliminated all problematic measurements. The final cleaned dataset contains **438 rows** covering six experimental sessions with reactor lengths from 25 cm to 2 m and diameters of 40 mm and 50 mm.

Table 3.1: Data cleaning pipeline summary.

Stage	Action	Rows remaining
Raw data	12 sessions, all measurements	726
Stage 1	Exclude 4 non-comparable sessions	631
Stage 2	Incomplete sweeps + flagged measurements	604
Stage 3	Exclude 12.5 cm reactor	438
Stage 4	Post-processing filters	438

3.5 Evaluation Metrics

All models in this project are evaluated on the same held-out test set (20% of the cleaned dataset, 88 rows) using four standard regression metrics. Each metric captures a different aspect of prediction quality, and reporting all four gives a more complete picture than any single number would.

R² (coefficient of determination) measures how much of the variation in the target variable the model explains. A value of 1.0 means the model explains the data perfectly, while 0.0 means it does no better than simply predicting the mean. R² is the most common headline metric for regression tasks and makes it easy to compare models at a glance, but it does not say anything about the size of the prediction error in the original units.

MAE (mean absolute error) is the average size of the prediction error, expressed in the same units as the target (g/kWh for ozone yield). A MAE of 2.5 g/kWh means the model's predictions are, on average, off by 2.5 g/kWh. MAE treats all errors equally: a prediction that is off by 10 g/kWh contributes ten times more to the MAE than one that is off by 1 g/kWh.

RMSE (root mean squared error) is also expressed in the units of the target, but unlike MAE it gives more weight to large errors because each error is squared before being averaged. This makes RMSE useful for spotting models that are usually accurate but occasionally produce very large mistakes. When RMSE is much larger than MAE, it signals that a few bad predictions are dragging the average error up; when they are close, the errors are evenly distributed.

MAPE (mean absolute percentage error) is the average size of the prediction error expressed as a percentage of the actual value. It makes errors easier to interpret across targets with different scales (for example, comparing a model predicting yield in g/kWh with one predicting production rate in g/h), since 3% means the same thing in both cases. MAPE can become unstable when the actual values are close to zero, but this is not an issue here because ozone yield in the dataset never falls below about 40 g/kWh.

Reporting all four metrics together is the standard approach in machine learning and provides a more balanced view of model performance than any one metric alone. R² gives a quick overall score, MAE gives an intuitive error size, RMSE highlights whether the model has any large failures, and MAPE allows comparison across different targets.

3.6 Model Development

This section describes the machine learning models trained to predict ozone yield from the six controllable input features. All models use the same 80/20 train-test split (350 training rows, 88 test rows, random state 42) and are evaluated on the same held-out test set to ensure a fair comparison. The modelling strategy follows a progressive complexity approach: starting with a linear baseline to establish a performance floor, then moving to tree-based ensemble methods that can capture non-linear feature interactions, and finally to neural network architectures including a physics-informed variant that incorporates domain knowledge as additional training signals.

3.6.1 Baseline Models

A Linear Regression model was trained as a baseline to establish a lower bound on predictive performance before introducing more complex methods. Linear Regression assumes a purely additive, linear relationship between each input feature and the target variable, making it the simplest possible model for this dataset.

On the held-out test set the model achieved $R^2 = 0.570$, $\text{MAE} = 9.91 \text{ g/kWh}$, and $\text{RMSE} = 12.59 \text{ g/kWh}$ ($\text{MAPE} = 13.0\%$). The low R^2 confirms that a linear model explains only about 57% of the variance in ozone yield, which is expected given the non-linear relationships described in the literature review: voltage has a threshold effect linked to the secondary streamer phase (non-monotonic in voltage), and ozone yield decreases non-linearly with energy density. A linear model cannot capture these patterns.

Despite its limited accuracy, the baseline serves two purposes. First, it confirms that the input features carry meaningful predictive signal, since even a linear fit explains over half the variance. Second, it provides a benchmark against which the more advanced models described in the following subsections can be compared.

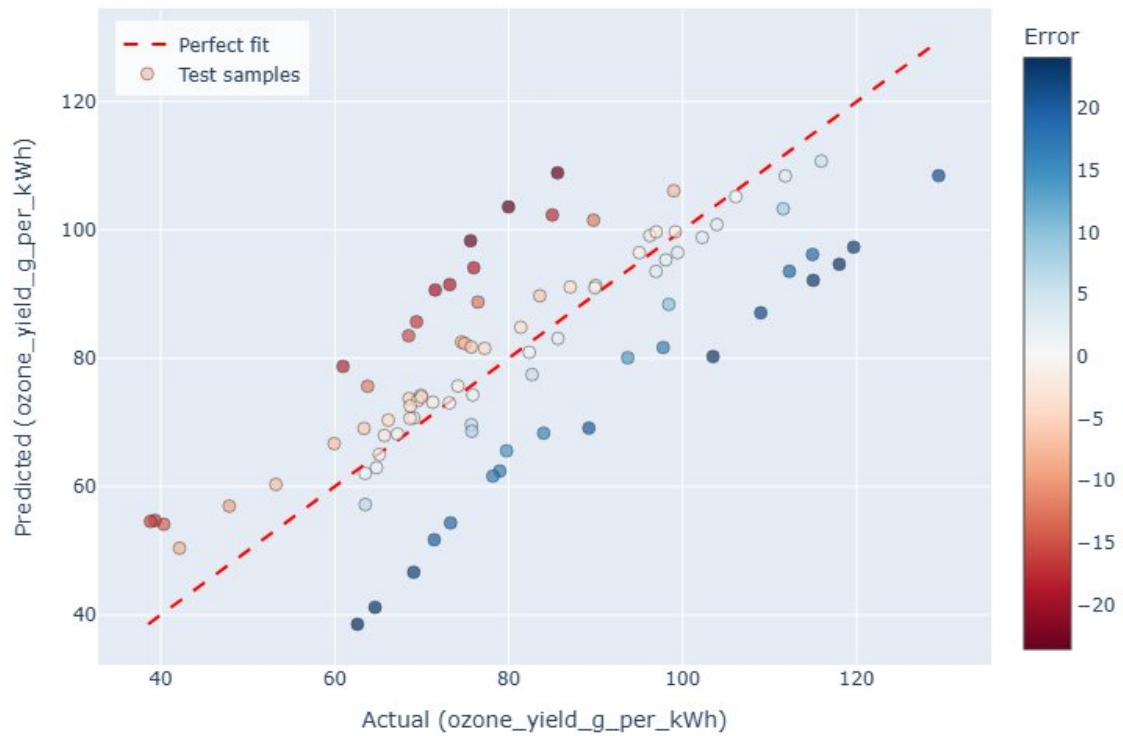


Figure 3.5: Linear Regression: predicted vs. actual ozone yield on the held-out test set. The red dashed line indicates the ideal one-to-one fit. The systematic spread around the diagonal, particularly at high yield values, illustrates the baseline model's inability to capture the non-linear relationships in the data.

3.6.2 Gradient Boosting

Gradient Boosting, first described by Friedman (2001), was selected as the starting point for this project because tree-based ensembles are a well-established default for small tabular datasets with non-linear relationships. In a large benchmark study across many tabular datasets, Grinsztajn et al. (2022) found that tree-based models such as gradient boosting remain highly competitive with, and often outperform, neural networks on this type of data, while being faster and easier to tune. This made gradient boosting a natural baseline to establish before moving on to the neural network models, whose performance on this specific dataset is compared against it in Chapter 4. The algorithm builds a collection of shallow decision trees sequentially, where each new tree is trained to correct the errors left by the previous trees. This iterative error-correction process allows the model to capture complex feature interactions without requiring additional feature engineering.

The model was trained using scikit-learn's `GradientBoostingRegressor` with 200 boosting rounds, a learning rate of 0.05, and a maximum tree depth of 4. The learning rate is a unitless scaling factor between 0 and 1 that controls how aggressively each tree corrects the previous predictions: each new tree's contribution is multiplied by this value before being added to the running total, so a lower value means more conservative updates, which generally improves generalisation at the cost of requiring more boosting rounds. A tree depth of 4 allows the model to capture interactions between up to four features in a single split path, which is sufficient to model relationships such as the connection between voltage, repetition rate, pulse width, and reactor geometry described in Section 2.1.

On the held-out test set the model achieved $R^2 = 0.968$, $\text{MAE} = 2.50 \text{ g/kWh}$, and $\text{RMSE} = 3.45 \text{ g/kWh}$ ($\text{MAPE} = 3.5\%$). Compared to the Linear Regression baseline ($R^2 = 0.570$), this represents a large improvement, confirming that the non-linear patterns in the data carry significant predictive information that a linear model cannot access.

Feature importance analysis showed that reactor length (32.4%) and repetition rate (26.9%) were the two most influential features, followed by voltage (13.6%), diameter (13.4%), flow rate (9.0%), and pulse width (4.7%). The dominance of reactor length and repetition rate is consistent with the physics: reactor length determines the active plasma volume and impedance matching, while repetition rate directly controls energy density without changing the plasma behaviour within each individual pulse.

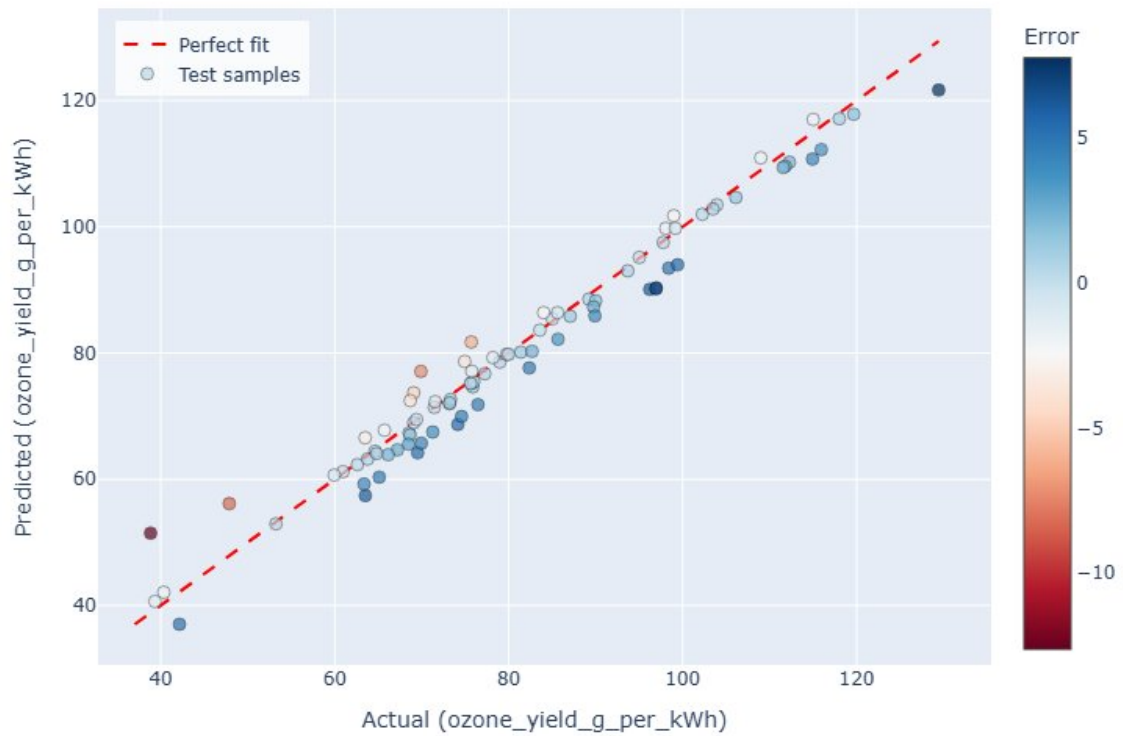


Figure 3.6: Gradient Boosting: predicted vs. actual ozone yield on the held-out test set. Predictions cluster tightly around the one-to-one line across the full yield range, showing that the tree-based model captures the non-linear feature interactions that the linear baseline misses.

3.6.3 XGBoost

XGBoost (Extreme Gradient Boosting), introduced by [Chen and Guestrin \(2016\)](#), follows the same sequential tree-building principle as the scikit-learn Gradient Boosting model described above, but uses a more optimised implementation with additional regularisation options. Two key differences are relevant here: XGBoost supports row and column subsampling per tree (similar to Random Forest), which helps reduce overfitting on small datasets, and it uses a more efficient split-finding algorithm that speeds up training.

The model was trained using `XGBRegressor` with the same core settings as the Gradient Boosting model (200 rounds, learning rate 0.05, max depth 4) plus two additional regularisation parameters: `subsample = 0.8` (each tree sees a random 80% of the training rows) and `colsample_bytree = 0.8` (each tree sees a random 80% of the features). These parameters introduce controlled randomness into the training process, reducing the risk of overfitting to any particular subset of the data.

On the held-out test set XGBoost achieved $R^2 = 0.976$, $\text{MAE} = 2.20$ g/kWh, and $\text{RMSE} = 2.97$ g/kWh ($\text{MAPE} = 2.9\%$), a small improvement over the standard Gradient Boosting model ($R^2 = 0.968$, $\text{RMSE} = 3.45$ g/kWh). The feature importance ranking differed slightly: reactor length remained the most important feature (30.7%), but flow rate (23.4%) and diameter (20.7%) ranked higher than in the Gradient Boosting model, while repetition rate (14.5%) and voltage (8.0%) ranked lower. Pulse width remained the least important feature (2.8%), the same position it had in the Gradient Boosting model. Reactor length staying on top in both models suggests it is a genuinely strong predictor, while the reranking of the remaining features is likely a combination of the regularisation introduced by row and column subsampling and the different split-finding algorithm XGBoost uses internally. With only 438 rows, small changes in which features each tree splits on can shift the importance values without changing prediction quality much.

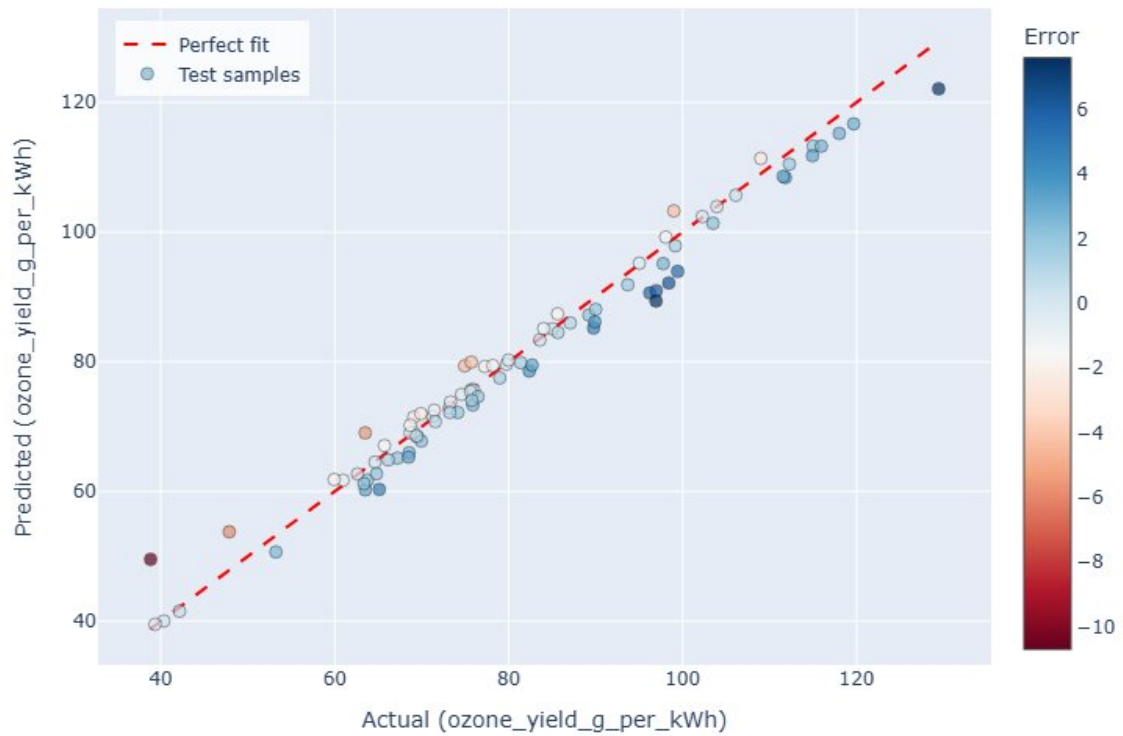


Figure 3.7: XGBoost: predicted vs. actual ozone yield on the held-out test set. The tighter clustering around the diagonal compared to the standard Gradient Boosting model (Figure 3.6) reflects the small accuracy gain from row and column subsampling.

3.6.4 Neural Network

A feedforward neural network was trained as the first deep learning model for this task. Neural networks can learn complex non-linear relationships between inputs and outputs, making them well suited for capturing the feature interactions that the tree-based models handle through sequential splitting.

The network architecture consists of three hidden layers with 128, 64, and 32 neurons respectively, each using the ReLU activation function, followed by a single linear output neuron. This architecture (each layer has half as many neurons as the one before it) is a standard pattern for small tabular regression tasks: the wider first layer gives the network enough capacity to learn useful combinations of the six input features, while the smaller later layers compress these combinations into a smaller set of higher-level patterns that map to the target. The layer sizes were selected through a systematic hyperparameter search (described later in this chapter), where $128 \times 64 \times 32$ was one of the top-performing baseline architectures without physics penalties. It was chosen as the production neural network for two reasons. First, among the top baselines it has the smallest number of parameters, which reduces the risk of overfitting on a dataset of only 438 rows. Second, the same architecture is also used in the Physics-Informed Neural Network described later in this chapter, so keeping the network size constant for both models isolates the effect of the physics penalties from the effect of changing the architecture.

Both the input features and the target variable were standardised using `StandardScaler` before training, which is standard practice for neural networks to ensure stable gradient updates across features with different scales. The model was trained using the Adam optimiser with a learning rate of 0.0005, a batch size of 32, and early stopping on a 15% validation split (patience = 20 epochs) to prevent overfitting.

To reduce the effect of random weight initialisation, five copies of the network were trained with different random seeds (42, 123, 456, 789, 1024) and their predictions were averaged to form a 5-seed ensemble. This averaging reduces variance in the predictions and produces more stable results than any single run.

The ensemble achieved $R^2 = 0.987$, $\text{MAE} = 1.58$ g/kWh, and $\text{RMSE} = 2.21$ g/kWh ($\text{MAPE} = 2.0\%$) on the held-out test set, improving over both Gradient Boosting ($R^2 = 0.968$) and XGBoost ($R^2 = 0.976$).

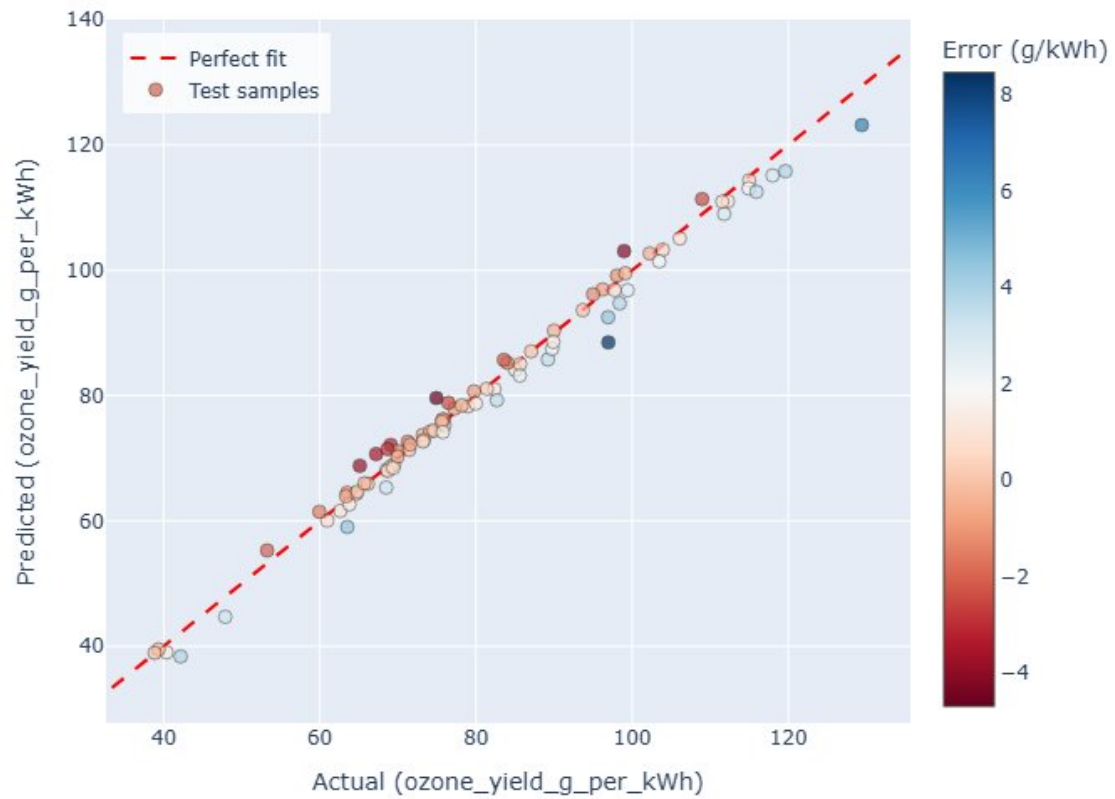


Figure 3.8: Neural Network 5-seed ensemble: predicted vs. actual ozone yield on the held-out test set. Predictions closely follow the one-to-one line across the full yield range, with the remaining deviations concentrated in the mid-yield region where the data density is highest.

3.6.5 Physics-Informed Neural Network

The standard neural network described above is data-driven: it learns whatever patterns reduce prediction error, regardless of whether those patterns make physical sense. To address this, a Physics-Informed Neural Network (PINN) was developed that adds monotonicity penalties to the training loss, encouraging the model to respect known physical relationships between the input features and ozone yield. The general PINN approach was introduced by [Raissi et al. \(2019\)](#) as a way to embed physical laws directly into the training objective of a neural network, originally for solving partial differential equations. The version used in this project follows the same idea (adding a physics-based penalty to the data-fit loss) but expresses the physics as monotonicity constraints rather than a differential equation, since the underlying plasma chemistry is not available in a closed-form equation that could be evaluated at every training point.

The monotonicity directions were derived from the experimental data and the plasma ozone literature discussed in Chapter 2:

- Voltage $\uparrow \Rightarrow$ yield $\uparrow$ (stronger electric field produces more energetic electrons, splitting O_2 molecules per pulse)
- Reactor length $\uparrow \Rightarrow$ yield $\downarrow$ (ozone breaks down in longer tubes, reducing efficiency)
- Repetition rate $\uparrow \Rightarrow$ yield $\downarrow$ (each additional pulse produces less additional ozone)
- Flow rate $\uparrow \Rightarrow$ yield $\uparrow$ (lower energy density per unit gas, keeping the reactor in the efficient operating range)

Physics-informed neural networks have already been used in plasma research: [Zhong et al. \(2022\)](#) demonstrate two PINN frameworks for low-temperature plasma simulation, using physical conservation laws as additional training signals. The PINN used in this project is simpler in scope (it works directly on the engineering parameters of the reactor rather than the full plasma equations), but follows the same general principle of using physical knowledge as a regulariser during training.

The PINN was implemented as a custom Keras model class (`PINNModel`) that extends the standard training loop. During each training step the model computes the standard mean squared error on the data, plus a physics penalty that is activated whenever the model's predicted direction for a feature goes against the expected direction listed above. Two parameters control the penalty: the physics weight w , which scales the penalty relative to the data loss, and delta δ , which defines a tolerance zone where small deviations are not penalised.

The PINN uses the same $128 \times 64 \times 32$ architecture and 5-seed averaging as the standard neural network, with all four monotonicity penalties active, $w = 0.05$, and $\delta = 0.15$. On the held-out test set it achieved $R^2 = 0.988$, $\text{MAE} = 1.48$ g/kWh, and $\text{RMSE} = 2.13$ g/kWh ($\text{MAPE} = 1.9\%$), a small but consistent improvement over the standard neural network across every metric (MAE , RMSE , MAPE , and median absolute error). This shows that the physics penalties not only aligned the model with known physical relationships but also produced slightly better generalisation to the held-out data: the physics signal acts as a form of regularisation that discourages the network from fitting noise in directions that contradict physics.

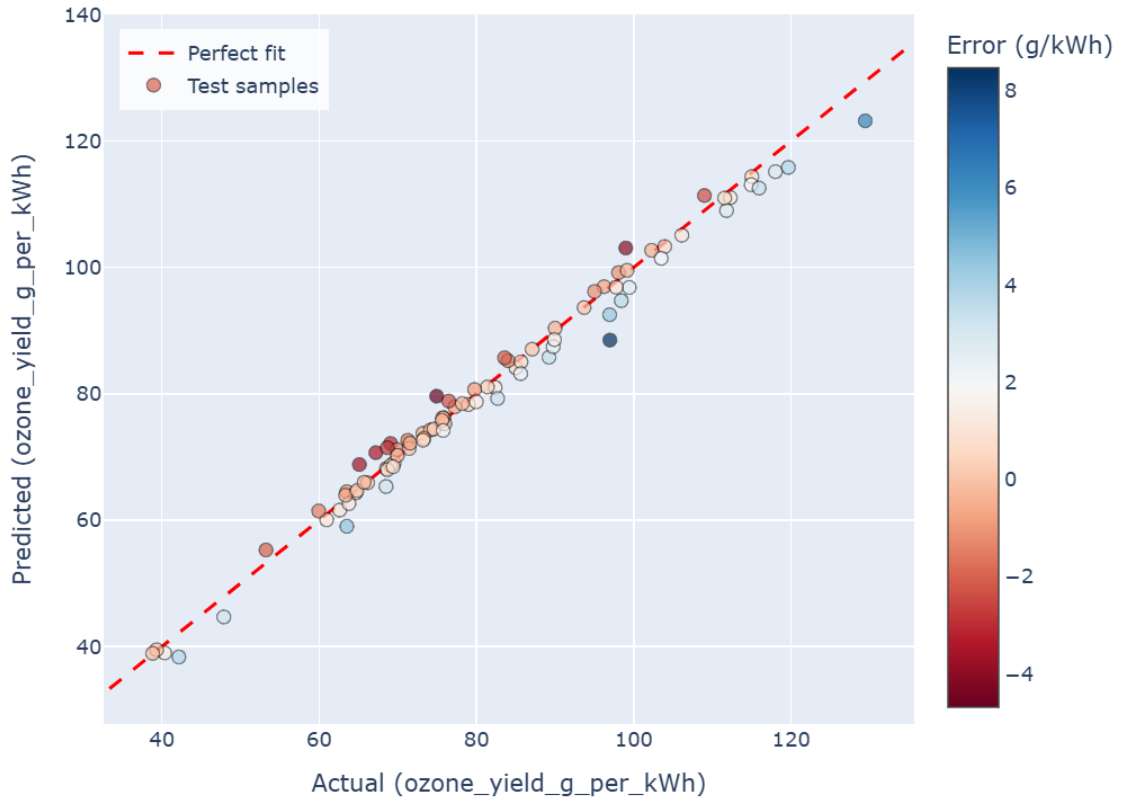


Figure 3.9: Physics-Informed Neural Network 5-seed ensemble: predicted vs. actual ozone yield on the held-out test set. The scatter pattern is visually comparable to the standard Neural Network (Figure 3.8) but with slightly tighter clustering around the diagonal, reflecting the small accuracy gain from adding physics-based regularisation.

3.6.6 Hyperparameter Tuning

Finding the best PINN configuration required a systematic search across both the network architecture and the physics penalty settings. This search was carried out in two phases, testing a total of 890 configurations.

Phase 1: Baseline Search (no physics)

Phase 1 tested 90 configurations without any physics penalties to find the best-performing network architecture and training settings. The search covered seven different layer architectures (ranging from $64 \times 32 \times 16$ to $256 \times 128 \times 64 \times 32$), three learning rates (0.001, 0.0005, 0.0003), three batch sizes (16, 32, 64), two alternative activation functions (ELU, Swish), L2 regularisation, dropout, and learning rate scheduling.

The top three baselines were:

1. $256 \times 128 \times 64$, lr = 0.0005, batch size 32 ($R^2 = 0.9868$)
2. $64 \times 32 \times 16$, lr = 0.001, batch size 16 ($R^2 = 0.9856$)
3. $128 \times 64 \times 32$, lr = 0.0005, batch size 32 ($R^2 = 0.9856$)

These three architectures were carried forward to Phase 2, together with the $128 \times 64 \times 32$ configuration at the higher learning rate (lr = 0.001, batch size 32) used in the standard neural network, giving four base networks for the physics search.

Phase 2: Physics Search

Phase 2 applied physics penalties to the four base networks from Phase 1. Nine different physics combinations were tested, ranging from single-feature penalties (e.g. voltage only, or flow rate only) to multi-feature combinations (e.g. voltage + flow rate, length + repetition rate) and the full set of all four penalties together (`all_physics`). Each combination was tested across eight physics weight values ($w = 0.01$ to 5.0) and five delta values ($\delta = 0.05$ to 0.3), producing 800 additional configurations.

All 20 top-performing configurations from the combined search used physics penalties, meaning every physics-based model outperformed the best baseline without physics ($R^2 = 0.9868$). The results across the top 20 were very close to each other, with R^2 ranging from 0.9884 to 0.9869 and RMSE between 2.07 and 2.20 g/kWh, showing that once physics penalties are added, the model performs well regardless of which specific combination is used. The single best configuration used only a voltage penalty with the smallest architecture ($64 \times 32 \times 16$, $w = 5.0$, $\delta = 0.2$, $R^2 = 0.9884$). However, looking at the top 20 results shows that different types of physics penalties all contributed to improving the model: voltage appeared in 7 of the top 20, voltage + flow rate in 4,

length + repetition rate in 1, and various other combinations in the rest. Importantly, the `all_physics` configuration (all four penalties active) appeared 5 times in the top 20 (ranks 9, 10, 12, 15, and 20), showing that using all physics penalties together is competitive with using individual penalties.

The final model used in production uses the $128 \times 64 \times 32$ architecture with all four physics penalties ($w = 0.05$, $\delta = 0.15$). This configuration was chosen over the single best scoring model for two reasons. First, applying all four penalties gives the model the broadest physical guidance, which is more useful when the model is asked to predict across the full range of reactor parameters. Second, while individual penalties like voltage-only scored slightly higher on R^2 , they only correct one relationship and leave the others unchecked.

3.6.7 Reactor Length Exclusion

The exclusion of the 12.5 cm reactor (Section 3.4, Stage 3) was validated by comparing model performance with and without the 12.5 cm data. Figure 3.10 shows this comparison for the Gradient Boosting model: including the 12.5 cm session drops R^2 from 0.971 to 0.941, with RMSE increasing from 3.29 to 6.12 g/kWh. The largest prediction errors are concentrated on the 12.5 cm data points, confirming that these measurements introduce noise the model cannot reliably learn from.

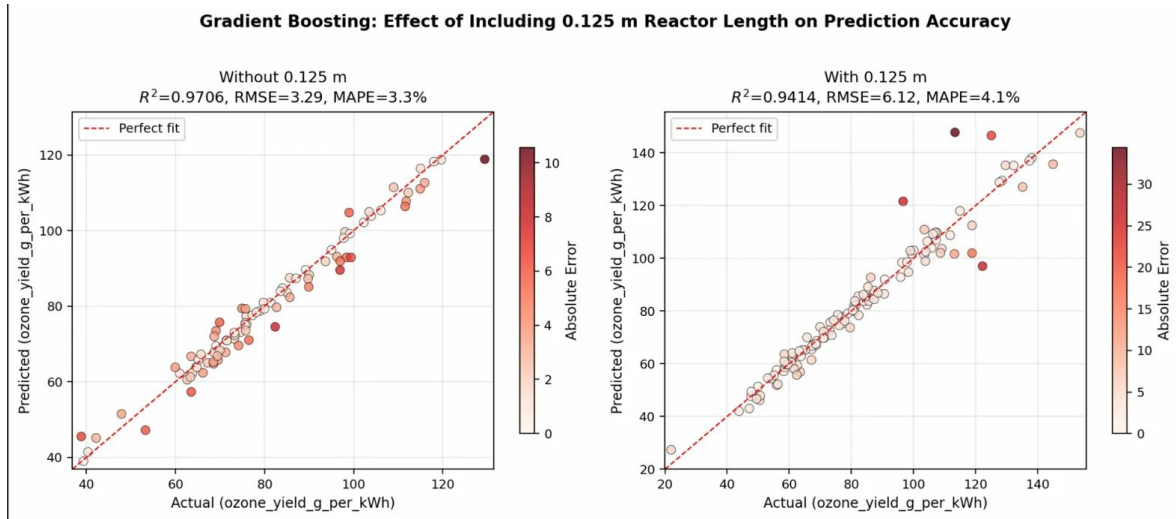


Figure 3.10: Effect of including the 12.5 cm reactor session on Gradient Boosting prediction accuracy. Left: model trained without the 12.5 cm session ($R^2 = 0.971$, RMSE = 3.29 g/kWh, MAPE = 3.3%). Right: model trained with the 12.5 cm session included ($R^2 = 0.941$, RMSE = 6.12 g/kWh, MAPE = 4.1%). The large errors visible in the right panel are concentrated on the 12.5 cm data points, supporting the decision to exclude this session.

3.7 Inverse Optimisation Framework

The models described in Section 3.6 solve the *forward* problem: given a set of reactor parameters, predict the resulting ozone yield. In practice, however, reactor operators face the opposite question: given a desired ozone yield (or production rate), what parameter settings should be used to achieve it? This is the *inverse* problem, and because the forward model is a non-linear function with no closed-form inverse, it must be solved numerically by searching the parameter space for the settings that bring the model's prediction closest to the desired target.

3.7.1 Problem Formulation

The inverse problem is framed as a bounded minimisation. The user selects a target variable (one of the four ozone metrics listed in Section 3.3) and a desired value for that variable. Each of the six controllable input parameters is then assigned one of two roles:

- **Fixed** parameters are held at a user-specified value. These represent constraints the operator cannot or does not want to change (e.g. a fixed reactor geometry).
- **Free** parameters are the variables the optimiser is allowed to search over. These are the values the operator want to adjust to hit the target.

The optimiser's job is to find the combination of free parameter values that, when passed through the trained forward model together with the fixed parameters, produces a prediction as close as possible to the user's desired target value. It does this by repeatedly trying different combinations of free parameter values, scoring each combination by how far its predicted ozone yield is from the target, and gradually converging on the combination with the smallest gap. The search is restricted to a bounded region: by default, each free parameter is constrained to the minimum and maximum values observed in the training data, which prevents the optimiser from returning solutions in unphysical or untrained regions of the parameter space. The user can optionally change these bounds further to encode additional operational constraints, such as a maximum voltage rating of the pulse source or a minimum flow rate required by the downstream process.

3.7.2 Optimisation Algorithm

To find the best combination of free parameter values, the framework uses an algorithm called differential evolution ([Storn and Price, 1997](#)), available in the `scipy.optimize` library. The idea behind it is straightforward: the algorithm starts with a group of random candidate solutions, where each candidate is one possible set of free parameter values. At every step, it mixes these candidates together to create new ones, checks which new candidates produce predictions closer to the target, and keeps the better ones while throwing out the worse ones. After many steps, the group of candidates converges on the combination of values that gets closest to the desired target.

This type of algorithm was chosen instead of a simpler method like gradient descent for two main reasons. First, gradient descent relies on a smooth response surface to know which direction to move in, but the Gradient Boosting model used here is built from decision trees, which produces a surface made of flat steps rather than a smooth slope. With no slope to follow, gradient descent cannot make progress. Second, the relationship between voltage and ozone yield is not a simple increasing or decreasing curve ([Chapter 2](#)), which means the response surface can have multiple low points. Gradient descent would just settle into the first low point it finds, while differential evolution keeps exploring and usually finds a better one. Both of these properties (non-differentiable response surfaces and multimodal objective functions) are well-known cases where evolutionary algorithms such as differential evolution are preferred over gradient-based methods, as discussed in the optimisation literature ([Das and Suganthan, 2011](#); [Storn and Price, 1997](#)).

The algorithm was configured with a population of ten candidates per free parameter, a maximum of 100 iterations, and a fixed random seed so that the same inputs always produce the same result.

3.7.3 Output and Interpretation

Once the optimiser finishes, it returns the combination of free parameter values that got closest to the desired target, along with the model's predicted value at that combination. The gap between the prediction and the target tells the user how well the target can be reached under the given constraints. A gap near zero means a valid set of parameters was found. A larger gap means the target cannot be reached within the chosen bounds, for example when the desired ozone yield is higher than what the reactor geometry can physically produce at any voltage or repetition rate.

This gap is reported to the user alongside the forward model's test-set MAE and MAPE, so that the accuracy of the recommended settings can be judged in context. A small gap combined with a low model error means the recommended settings are trustworthy. A small gap combined with a high model error means the optimiser found parameters that match the target according to the model, but the model itself may not be accurate enough at those settings for the recommendation to be reliable.

4 — Results and Discussion

4.1 Predictive Model Performance

This section compares all five models trained in Chapter 3 on the same held-out test set of 88 measurements. Each model was given the same six controllable input features and the same train/test split, so the only difference between them is the type of model and how it learns.

Table 4.1 reports the four headline metrics (R^2 , MAE, RMSE, MAPE) plus two additional metrics that describe the spread of errors (Max Error and Median Absolute Error). The numbers are taken directly from the per-model results in Sections 3.6.1 to 3.6.5.

Table 4.1: Test set performance for all five models. MAE, RMSE, Max Error and Median AE are reported in g/kWh. MAPE is reported in percent. The best value in each column is in bold.

Model	R^2	MAE	RMSE	MAPE	Max Error	Median AE
Linear Regression	0.570	9.91	12.59	13.0%	24.07	7.11
Gradient Boosting	0.968	2.50	3.45	3.5%	12.63	1.73
XGBoost	0.976	2.20	2.97	2.9%	10.71	1.84
Neural Network	0.987	1.58	2.21	2.0%	8.08	1.00
PINN	0.988	1.48	2.13	1.9%	9.98	0.92

The headline result is that the PINN performs best on five of the six metrics, with $R^2 = 0.988$, MAE = 1.48 g/kWh, RMSE = 2.13 g/kWh, and MAPE = 1.9%. Figure 4.1 visualises the R^2 ranking from the table, and Figure 4.2 shows the three error metrics side by side.

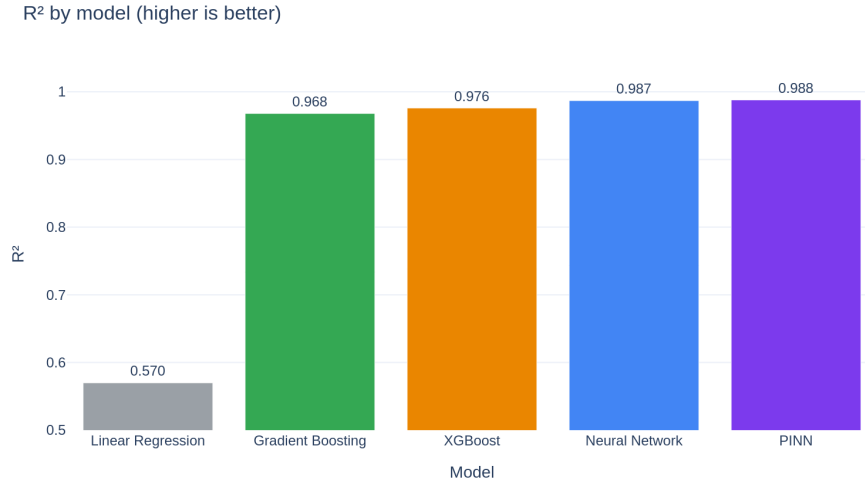


Figure 4.1: R^2 on the unseen test set for all five models. Linear Regression explains only 57% of the variance, while the four non-linear models all reach $R^2 \geq 0.968$.

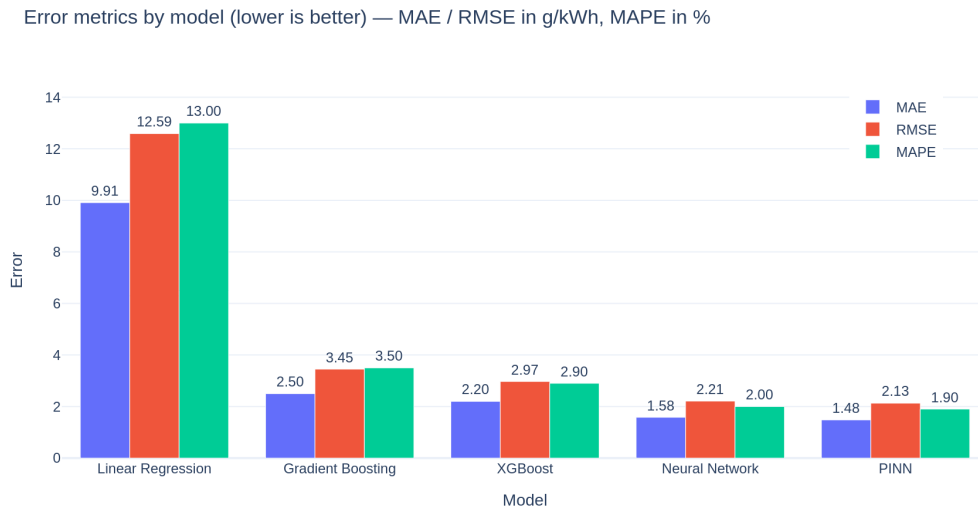


Figure 4.2: Mean Absolute Error, Root Mean Squared Error, and Mean Absolute Percentage Error for each model on the unseen test set. MAE and RMSE are in g/kWh, MAPE is in percent.

The progression from Linear Regression to the PINN tells a clear story. The Linear Regression baseline explains only 57% of the variance in ozone yield, with an average error of about 10 g/kWh. This confirms the prediction made in the literature review: a linear model cannot capture the non-linear effects that drive plasma ozone production, in particular the threshold behaviour of voltage and the inverse relationship between energy density and yield.

Switching to Gradient Boosting closes most of this gap in a single step. R^2 jumps from 0.570 to 0.968 and MAE drops by a factor of four, from 9.91 to 2.50 g/kWh. The

reason is straightforward: tree-based models can split on different features at different operating points, which lets them represent the same kind of “if voltage is in this range, then...” rules that the underlying physics produces. XGBoost adds row and column subsampling on top of the same boosting principle and gains another small improvement, reaching $R^2 = 0.976$ and $\text{MAE} = 2.20$ g/kWh.

The two neural network models close the remaining gap. The plain Neural Network reaches $R^2 = 0.987$ with $\text{MAE} = 1.58$ g/kWh, and the PINN improves on every central tendency metric, reaching $R^2 = 0.988$ with $\text{MAE} = 1.48$ g/kWh. The improvement from XGBoost to the Neural Network is larger than the improvement from the Neural Network to the PINN, which suggests that most of the remaining accuracy gain comes from the network’s ability to learn smooth feature interactions, while the physics penalties contribute a smaller additional refinement.

Figure 4.3 shows how the prediction errors are spread for each model, using the four reported error statistics: Median AE, MAE, RMSE, and Max Error. For each model, the vertical spine connects the smallest of these values (the typical error) to the largest (the worst single prediction), so the height of each spine is a quick visual measure of how consistent the model is.

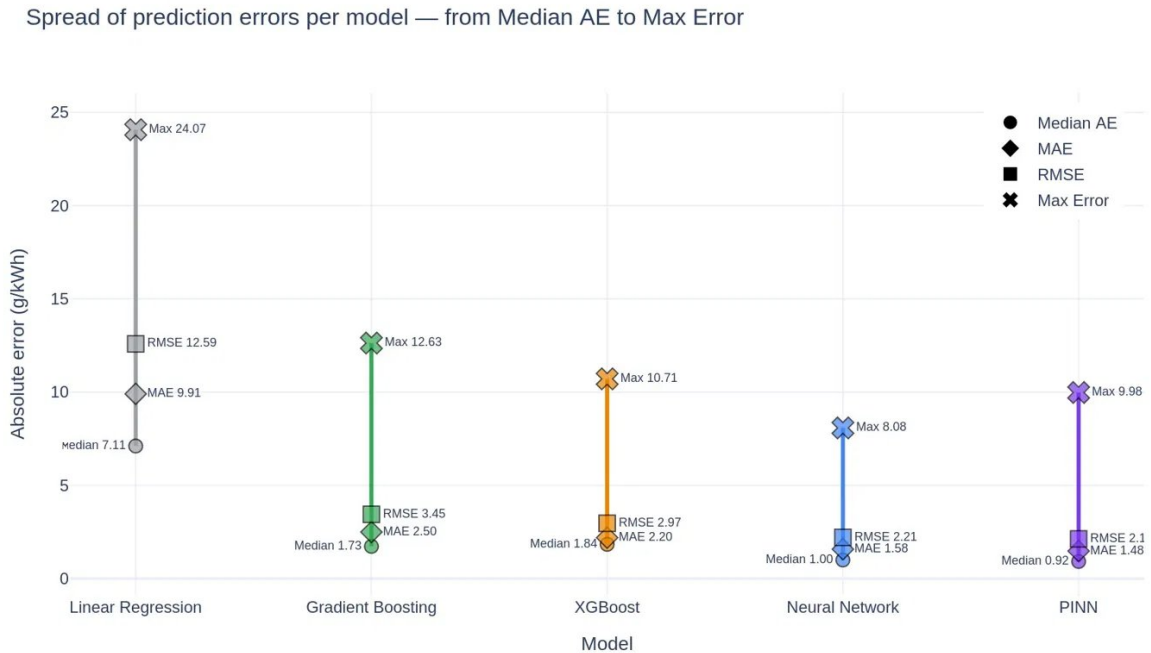


Figure 4.3: Spread of prediction errors per model. Each spine connects the four reported error statistics (Median AE, MAE, RMSE, Max Error) so that both the typical error and the worst-case error are visible at a glance.

Two patterns stand out in this figure. First, Linear Regression has by far the longest spine, with a median error of 7.11 g/kWh and a maximum of 24.07 g/kWh. Even on a typical test point, the linear model is off by more than the maximum error of the Neural Network. Second, all four non-linear models share a similar shape: their Median AE, MAE and RMSE cluster tightly together near the bottom of the chart, while the Max Error sits well above this cluster. This shape means the models are accurate most of the time and only occasionally produce a noticeably large error, which is consistent with the test set covering multiple reactor configurations and operating conditions.

A closer look at the four non-linear models also shows that the Neural Network and the PINN each win on different metrics. The PINN wins on every central tendency metric (Median AE = 0.92 vs 1.00, MAE = 1.48 vs 1.58, RMSE = 2.13 vs 2.21), but the Neural Network has the lowest Max Error of any model in the comparison (8.08 vs 9.98 g/kWh). In other words, the physics penalties pull the typical error down but do not necessarily improve the worst-case prediction. For the inverse optimisation use case described in Section 3.7, the central tendency metrics matter more than the worst-case error, because the optimiser averages over many forward predictions during its search.

A note on the size of the differences between the four strong models. With only 88 rows in the unseen test set, small absolute gaps in MAE (such as 0.10 g/kWh between the Neural Network and the PINN) correspond to only a handful of test points predicted slightly differently. The R^2 values for the top three models also fall within a narrow band (0.968 to 0.988), which means the ranking between Gradient Boosting, XGBoost, the Neural Network, and the PINN should be read as “all four are good, with the PINN slightly ahead” rather than as evidence of a large gap. The choice between them in practice depends on factors beyond accuracy, such as training time, ease of deployment, and how naturally the model can be combined with physical knowledge.

In summary, all four non-linear models achieve test set accuracy good enough for the prediction tool to be useful in practice (MAE below 3 g/kWh, MAPE below 4%), and the PINN is the most accurate of the five on the metrics that matter most for the inverse optimisation framework. The small but consistent improvement from the Neural Network to the PINN is consistent with what [Zhong et al. \(2022\)](#) report for physics-informed neural networks in low-temperature plasma simulation, where adding a physics-based loss term provides a measurable benefit over a purely data-driven baseline. The remaining sections of this chapter examine which features drive these predictions (Section 4.2), how well the inverse optimisation framework recovers reactor settings for a given target ozone output (Section 4.3)

4.2 Feature Importance

Understanding which features the trained models rely on most matters for three reasons. First, it is a check on the model itself: if a feature ranks much higher or lower than the underlying physics would suggest, it can point to overfitting, data issues, or a model that has learned a pattern that does not match reality. Second, it makes the predictions easier to trust and explain. Engineers at KapiteinLabs and external clients are more likely to use the dashboard when they can see which reactor parameters drive the predicted ozone yield and verify that those parameters line up with their physical intuition. Third, the ranking informs the inverse optimisation tool: knowing which features dominate the prediction tells the operator which parameters to focus on when planning new experiments.

The previous section showed that the four non-linear models all reach $R^2 \geq 0.968$ on the test set, but it did not say which features they actually rely on to make those predictions. Each model type also reports importance differently: tree-based models report the reduction in error each feature provides during training, neural networks have no built-in importance score at all, and Linear Regression reports a coefficient that is not directly comparable to either of the above. To compare all five models on the same scale, this section uses *permutation importance* on the test set.

Permutation importance works the same way for every model: it shuffles one feature’s values across the test rows, predicts on the shuffled data, and measures how much R^2 drops compared to the unshuffled baseline. The bigger the drop, the more the model relied on that feature. The shuffle is repeated 30 times per feature and averaged, so the result is not driven by a single lucky permutation. Because the same procedure is used for every model, the resulting numbers can be placed side by side without any further adjustment.

To put the five models on the same footing, the raw R^2 drops are normalised so that each model’s importance values sum to 1. This converts the result into a ”share of importance” per feature, which removes the effect of each model having a different baseline R^2 . The normalised values are reported in Table 4.2.

Table 4.2: Normalised permutation importance per model. Each row sums to 1 and represents the share of the model’s predictive signal that comes from each feature. The bottom row shows the average across the four non-linear models.

Model	Length	Diameter	Voltage	Rep. rate	Flow rate	Pulse width
Linear Regression	0.197	0.064	0.334	0.291	0.065	0.049
Gradient Boosting	0.402	0.071	0.097	0.335	0.056	0.039
XGBoost	0.388	0.115	0.102	0.340	0.039	0.016
Neural Network	0.350	0.079	0.203	0.231	0.108	0.029
PINN	0.351	0.085	0.188	0.231	0.104	0.040
Consensus (4 non-linear)	0.373	0.087	0.148	0.284	0.077	0.031

Figure 4.4 shows the consensus ranking from the bottom row of the table.

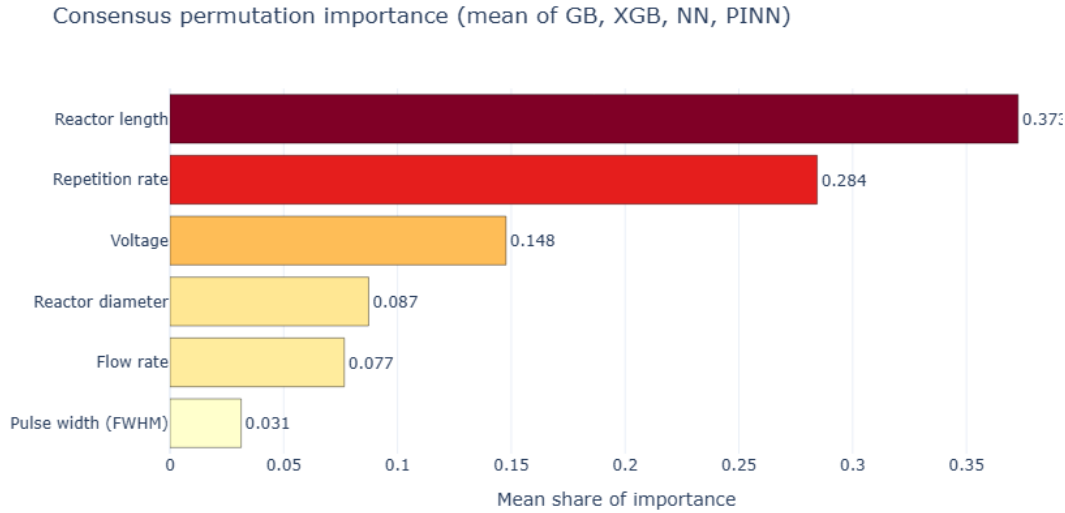


Figure 4.4: Consensus permutation importance, averaged across the four non-linear models (Gradient Boosting, XGBoost, Neural Network, PINN). Reactor length and repetition rate together account for roughly 66% of the total importance.

The two clearest results from Table 4.2 are at the top and the bottom of the consensus ranking. **Reactor length and repetition rate together account for about 66% of the predictive signal** across the four non-linear models, with reactor length first (37.3%) and repetition rate second (28.4%). Reactor length sets the active plasma volume and the impedance seen by the pulse source, while repetition rate directly controls energy density without changing the plasma behaviour within each pulse. **Pulse width is consistently last** for every model, with a share between 1.6% and 4.9%. This does not mean pulse width is physically unimportant. It means that within the range of pulse widths covered in Campaign 5 (0.09–0.2 μ s), variation in pulse width does not drive much variation in ozone yield once the other features are accounted for.

Two other patterns in the table are also useful to highlight. The first is how the models disagree on the importance of **voltage**. Linear Regression places voltage as its single most important feature (33.4%), while the four non-linear models all rank it third or lower, with the tree-based models giving it only about 10%. This difference is a useful check on the modelling approach. As described in Section 2.1.2, voltage has a threshold-shaped effect on ozone yield: once the voltage rises above the streamer ignition threshold, further increases push the reactor toward the inefficient secondary streamer phase and yield drops. A linear model has no way to represent this and tries to fit the threshold pattern as a straight slope, which inflates the apparent importance of voltage. The non-linear models can carve out the threshold behaviour and rely more

on the features that scale linearly with yield, such as reactor length and repetition rate.

The second pattern is the split between tree-based and neural models on **flow rate** and **voltage**. The two neural models give flow rate a 10–11% share and voltage about 19%, while the two tree-based models give flow rate only 4–6% and voltage about 10%. This is consistent with how the two model families learn: trees produce sharp splits along the dominant axes (length, repetition rate) and lean heavily on those, while neural networks distribute attention more smoothly across the input space and pick up smaller but real contributions from the remaining features. Both views are valid, and the consensus ranking averages over them.

One important point applies to all of these numbers. Permutation importance measures how much each feature contributes to model accuracy on the test set, within the range of values seen during training. It does not say anything about which settings are safe to operate at, or about which features can be left at any value without consequence. The implications of this distinction for the inverse optimisation tool and for reactor operators are discussed in Section 4.4.

4.3 Inverse Optimisation Results

This section evaluates how well the inverse optimisation framework described in Section 3.7 recovers reactor settings for a user-defined target ozone yield. The PINN was used for the optimiser, since it was the most accurate of the five models trained in Chapter 3. As described in Section 3.7.3, the quality of an optimisation result is judged by two numbers: the gap between the optimiser’s predicted output and the target, and the model’s own test set error. A small gap combined with a low model error means the recommended settings are trustworthy. The optimiser’s performance is evaluated here against the model’s predictions rather than against new reactor measurements; validating the recommended settings in the lab is left as future work.

Three example runs are reported in Table 4.3 to cover the main use cases. Run 1 asks for a mid-range yield with all six parameters free. Run 2 asks for a high yield near the upper edge of the dataset, also with all parameters free. Run 3 simulates the most common client scenario: the reactor geometry is fixed (1.0 m length, 50 mm diameter) and the optimiser is asked to find pulse and flow settings that reach a target yield.

In all three runs the optimiser hits the target almost exactly, with gaps between 0.001 and 0.006 g/kWh. These gaps are three orders of magnitude smaller than the PINN’s own MAE on the test set, which means the residual mismatch between the optimiser’s prediction and the target is very small compared to the model’s own uncertainty. The constrained run (Run 3) shows that the optimiser also handles the realistic client scenario well: with reactor geometry locked at 1.0 m by 50 mm, it still finds a combination of voltage, repetition rate, flow rate, and pulse width that reaches the

Table 4.3: Three example inverse optimisation runs on the PINN. “Free” parameters are searched by the optimiser; fixed parameters are user-specified constraints. The gap is the difference between the optimiser’s predicted output and the target yield.

	Run 1	Run 2	Run 3
Target (g/kWh)	80.00	110.00	70.00
Achieved (g/kWh)	80.00	110.00	70.01
Gap (g/kWh)	0.004	0.001	0.006
Reactor length (m)	1.309	0.451	1.000 (fixed)
Reactor diameter (mm)	47.79	48.73	50.0 (fixed)
Voltage (kV)	10.45	12.27	10.68
Repetition rate (kHz)	4.15	1.36	4.98
Flow rate (L/min)	19.01	17.73	12.91
Pulse width (μ s)	0.131	0.087	0.174

70 g/kWh target. The recommended settings in all three runs sit comfortably inside the trained parameter ranges.

A practical note on the role of pulse width is also worth making here. Pulse width has the lowest feature importance of all six inputs (3.1% in the consensus ranking from Section 4.2), but adding it as a sixth controllable parameter in the optimisation framework had a clear effect on the gap to target. Before pulse width was included, the optimiser frequently ended its search with a gap of around 1 g/kWh on the same kind of targets shown in Table 4.3. After adding pulse width, the gap dropped to consistently below 0.01 g/kWh across most runs. This shows that low feature importance does not mean low usefulness in optimisation: an extra degree of freedom helps the optimiser fine-tune the prediction, even if the feature itself contributes little to model accuracy.

These results answer the third research question of the project: within the trained parameter range, the inverse optimisation method can recommend reactor settings that produce a user-defined target ozone yield to a precision well within the model’s accuracy. The behaviour of the tool outside the trained range, the choice of model, and the implications for clients are discussed in Section 4.4.

4.4 Discussion

The results in the previous three sections answer the three research questions posed in Section 1.4.

The first sub-question asked how accurately a machine learning model can predict ozone yield from the controllable reactor parameters, and how different model families compare on this dataset. The four non-linear models all reach $R^2 \geq 0.968$ on the test set, with the PINN achieving the lowest error (MAE = 1.48 g/kWh, MAPE = 1.9%). The differences between the four non-linear models are small in absolute terms, with all of them producing test-set accuracy good enough for the prediction tool to be useful in practice. This level of accuracy is in line with what other recent ML-based plasma reactor work has reported. [Shi et al. \(2024\)](#) used a similar data-driven workflow on a thermal plasma reactor, and [Li et al. \(2025\)](#) reported that an artificial neural network predicted the performance of a dielectric barrier discharge reactor in close agreement with experiment. Both support the wider observation that machine learning can capture the non-linear behaviour of plasma systems well when trained on a structured dataset, and the results here extend that observation to a pulsed-plasma ozone reactor.

The second sub-question asked what impact adding physical knowledge in the form of monotonicity constraints has on prediction accuracy and physical consistency. Adding the four monotonicity penalties produced a small but consistent improvement on every central tendency metric (MAE, RMSE, MAPE, Median AE), and the resulting model is also more aligned with the physics of plasma ozone production.

The third sub-question asked how well an inverse optimisation method built on top of the predictive model can recommend reactor settings for a user-defined target ozone output within the trained parameter range. The framework recovers settings that hit user-defined yield targets to within 0.01 g/kWh in most cases, well below the model's own MAE of 1.48 g/kWh, and also handles realistic client scenarios with fixed reactor geometry while staying inside the trained parameter range.

Operational Risks

One important point applies to the feature importance results in Section 4.2: a feature's importance score does not say anything about its operational safety. The clearest example is voltage. The model treats voltage as a smooth input and predicts ozone yield smoothly down to any value the optimiser explores, including values well below the streamer ignition threshold described in Section 2.1.2. In a real reactor, dropping the voltage below this threshold prevents plasma ignition entirely and produces no ozone at all. The model has no way of learning this from the data, because the dataset only contains successful runs (10–15 kV). The same applies in different ways to repetition rate (too low at high voltage damages the pulse source), flow rate (too low overheats the reactor), and reactor length (too short prevents stable streamer formation). The optimisation tool should therefore be used with operator-defined bounds that reflect the physical and safety limits of the actual hardware, not just the trained range of the dataset.

Ethical Considerations

A few ethical points are worth making here, since the dashboard will be used by both internal engineers and external clients to support real design and purchasing decisions. The model's predictions are statistical estimates, not certainties: the dashboard always reports the model's MAE and MAPE alongside every prediction, and flags inputs that fall outside the trained range, so users can judge for themselves how much to trust a given result. The optimisation tool is intended to support engineering judgment rather than replace it; a recommended set of reactor settings should be treated as a starting point for further analysis and laboratory testing, not as a final design. The dataset used in this project was shared by TU Eindhoven with KapiteinLabs's awareness, contains no personal data, and does not involve human or animal subjects, so the main ethical responsibility for the work lies in honest reporting of the model's accuracy and its limitations. On the wider context, improving the efficiency of ozone-based treatment reduces the energy required to produce a given amount of ozone, which has a small but real environmental benefit in industrial water and air treatment applications.

Limitations

Three limitations are important enough to highlight here. First, the dataset covers a narrow part of the design space: six experimental sessions, two reactor diameters (40 and 50 mm), two flow rates (10 and 20 L/min), and dry synthetic air only. Predictions outside these ranges go beyond what the model was trained on, and should be treated as rough estimates rather than reliable answers. Second, the dataset was collected by an external research group at TU Eindhoven, so adding new measurements to expand the model's coverage depends on the schedule of those experiments. Third, environmental variables that are known to affect plasma behaviour (gas temperature, humidity) were not measured during Campaign 5. Adding these sensors in future experiments would let the model account for variation in operating conditions that it currently treats as constant. A separate direction for future research is validating the inverse optimisation framework in the lab: running a few of the recommended configurations on the actual reactor and comparing the measured ozone output to the predicted value would turn the optimisation tool from a model-based recommendation into a tool that engineers can rely on for real design decisions.

Despite these limitations, the deliverable answers the project's main research question. Within the trained parameter range, the prediction tool is accurate enough to inform experiment planning, and the optimisation tool reliably recovers reactor settings for a target ozone output. The recommendations are physically reasonable and the framework is designed to grow with new data: once new measurements arrive, the dashboard allows the underlying model to be retrained on the expanded dataset without code changes.

5 — Dashboard

5.1 Overview

The dashboard is the user-facing deliverable of this project. It packages the trained models and the inverse optimisation framework described in Chapter 3 into a single interface that engineers at KapiteinLabs and external clients can use without writing any code. The dashboard is built in Streamlit, a Python framework for interactive web applications, and is deployed using Streamlit Community Cloud. This means the tool can be opened in any web browser without local installation.

The interface is organised into a small number of clearly labelled areas. A header at the top shows the KapiteinLabs branding and identifies the dataset the models are trained on (currently Campaign 5). A Data Management section at the top of the page allows the user to upload new experimental data and retrain the underlying models on the expanded dataset. The main working area is split into two tabs: **Forward Prediction** for predicting ozone output from a chosen set of reactor settings, and **Inverse Optimization** for finding reactor settings that produce a target ozone output. The two tabs correspond to the two project deliverables introduced in Section 1.1. The visual style of the dashboard (colour palette, typography, and layout) follows the KapiteinLabs brand book, so the tool sits naturally alongside other internal and client-facing materials.

5.2 Forward Prediction

The Forward Prediction tab is shown in Figure 5.1. The user starts by choosing one of the four ozone targets defined in Section 3.3 (yield in g/kWh, concentration in g/m³, ppm, or production rate in g/h) and one of the four trained models. The model dropdown labels each model with a short plain-English tag: *Gradient Boosting – fast prediction*, *XGBoost – advanced fast prediction*, *Neural Network – complex prediction*, and *PINN – physics-powered prediction*. These tags help non-technical users pick a model without needing to know the underlying machine learning details.

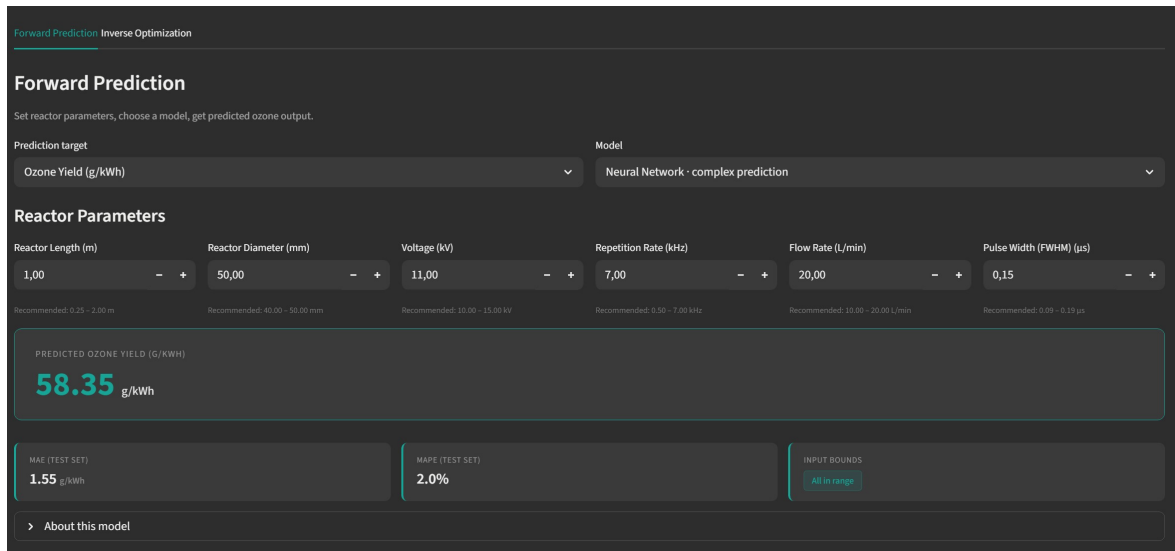


Figure 5.1: Forward Prediction interface. The user sets six reactor parameters, picks a target and a model, and the predicted ozone output is shown live in the result card. The MAE and MAPE of the chosen model on the test set are displayed, alongside an indicator showing whether all inputs are within the trained parameter range.

Below the model selector, six numeric input fields let the user set the reactor parameters: reactor length, reactor diameter, voltage, repetition rate, flow rate, and pulse width. Each field shows the recommended range below the input box, taken directly from the training data. As the user changes any parameter, the predicted ozone output updates live in a large result card at the bottom of the section. Two smaller cards next to it show the chosen model’s MAE and MAPE on the test set, so the user always sees the model’s accuracy alongside its prediction. A third card labelled *Input Bounds* shows whether all six inputs are within the trained range or if any of them sits outside it. This is the dashboard’s direct response to the operational risk discussion in Section 4.4: the user is reminded whenever their inputs leave the region where the model has been validated.

The Forward Prediction tab also includes an *About this model* panel that explains the chosen model in one short paragraph of plain English, accompanied by the predicted-vs-actual scatter plot from Section 4.1. The panel also includes explanations of MAE and MAPE so that a non-technical user can read the accuracy values without machine learning knowledge. A second collapsible panel, *MAE by reactor length*, breaks down the model’s accuracy per reactor length used in the training data. This gives the user a view of where the model performs well and where its errors are larger, which is useful when deciding how much to trust a prediction for a specific reactor geometry.

5.3 Inverse Optimisation

The Inverse Optimization tab implements the framework described in Section 3.7. Figure 5.2 shows the input section of the tab. The user begins by selecting a target metric, a desired value for that metric, and the model to use as the prediction engine inside the optimiser. A small text label below the desired value field shows the range of values seen in the training data, so the user can tell whether their chosen target is realistic or whether it lies outside the trained range.

Figure 5.2: Inverse Optimization input section. The user selects a target metric, a desired value, and a model, then assigns each of the six reactor parameters to either Optimize (free) or Fix (held at a user-specified value). The example shown sets a target yield of 70 g/kWh with reactor length and diameter fixed at 1 m and 50 mm.

The next section, *Parameter Roles*, lets the user set each of the six reactor parameters to one of two roles: **Optimize** (the parameter is free and the optimiser will search over it) or **Fix** (the parameter is held at a user-specified value and the optimiser leaves it alone). When a parameter is set to Fix, a numeric input appears below it for the fixed value. An additional checkbox labelled *Set custom search bounds* lets advanced users override the default search range for the free parameters, for example to encode a hardware limitation such as a maximum voltage rating.

Pressing **Run Optimization** starts the differential evolution search described in Section 3.7.2. The result is shown in a layout that mirrors the Forward Prediction tab so the user does not have to learn a second interface. Figure 5.3 shows the result of the optimisation run set up in Figure 5.2.

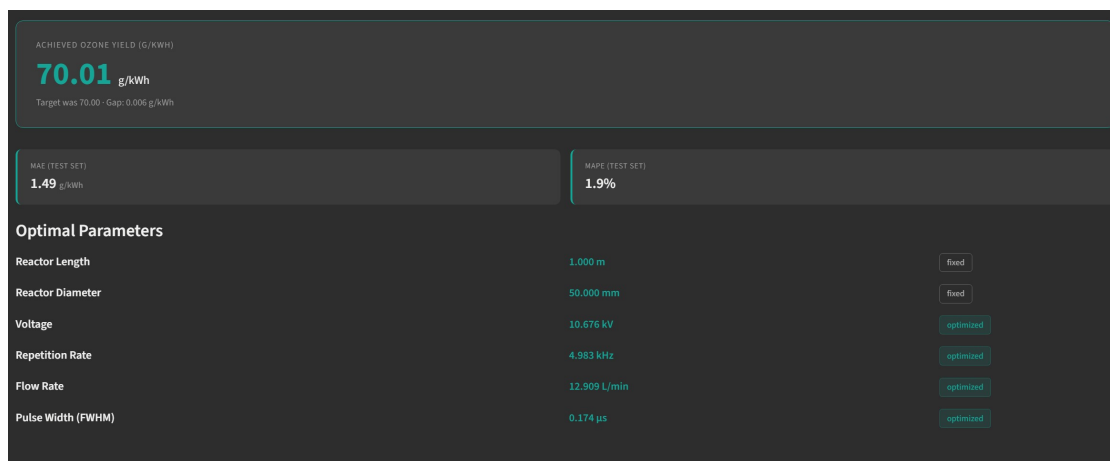


Figure 5.3: Inverse Optimization result for the run configured in Figure 5.2. The optimiser hit the target almost exactly (gap of 0.006 g/kWh) and reports the four free parameters it selected, alongside the model’s MAE and MAPE for context.

The result card at the top reports the achieved ozone output, the gap to the target, and the model’s MAE and MAPE. Below it, the recommended values for all six parameters are listed, with each one tagged as either *fixed* or *optimized*. This layout makes it immediately clear which numbers came from the user and which came from the optimiser, which is important when the user wants to share the result with a colleague or a client.

Together, the two tabs cover both directions of the prediction problem and give the user a single tool for either checking how a planned set of reactor settings will perform or finding the settings needed to hit a specific ozone target. The Data Management section at the top allows the dashboard to remain useful as more experimental data becomes available, by retraining the models on the expanded dataset without any code changes.

5.4 Testing and Evaluation

The dashboard was evaluated in two ways. The first is functional testing: a set of manual checks to confirm that all features work well. The second is a user evaluation: a short structured survey sent to people at KapiteinLabs and informal feedback gathered from an external researcher who works daily with pulsed plasma reactors, designed to capture how the tool feels from the perspective of its real audience.

Functional Testing

Before sending the dashboard out for review, every feature was checked manually on the deployed version at `kapiteinlabs-ozone.streamlit.app`. The four models load correctly and produce predictions within a fraction of a second. The Forward Prediction tab updates the predicted ozone yield live as inputs are changed, and the *Input Bounds* card correctly flags inputs that sit outside the trained parameter range. The Inverse Optimisation tab returns results that match the standalone notebook implementation: identical inputs produce identical recommendations and gaps, confirming that the Streamlit version of the optimiser is consistent with the development environment. Fixed parameters stay at their user-specified values throughout the search, and the custom search bounds option correctly restricts the optimiser to user-defined ranges.

User Evaluation

The structured evaluation form was completed by three reviewers at KapiteinLabs, covering three different roles: an intern working as an electrical engineer and data analyst, the Chief Scientific Officer, and a data scientist. In addition, informal feedback was gathered from an external researcher who builds and operates pulsed plasma reactors as part of their daily work and was directly involved in the Campaign 5 experiments. The three KapiteinLabs reviewers used a structured form with 1–5 rating questions and open comments, while the external feedback was collected as meeting notes. This mix gives both quantitative ratings and qualitative comments from a balanced audience: a technical leader, two analysts with different backgrounds, and a hands-on plasma reactor expert from outside the company.

The reviewers also varied in how much background they brought to the task. Self-reported familiarity with plasma reactor physics ranged from 2 to 5 (average 3.3), and machine learning experience ranged from 3 to 5 (average 4.0). This range is intentional: the dashboard is designed to be used by both technical specialists and non-specialists, so feedback from a mixed audience is more informative than feedback from a single user group.

Table 5.1 summarises the rating questions. The results are grouped by the three parts of the form: usability, scientific correctness, and overall judgment.

Table 5.1: Average user evaluation ratings (1 = strongly disagree, 5 = strongly agree). All usability and overall questions received the maximum score from every reviewer.

Question	Average
<i>Part 1 – Usability</i>	
Purpose of the dashboard was clear from the first minute	5.0
Setting the six reactor parameters was easy	5.0
Labels, units, and recommended ranges were clear	5.0
Inverse Optimisation tab was clear to use	5.0
Results were easy to read and understand	5.0
<i>About this model</i> panel was helpful	5.0
Dashboard responded quickly and felt stable	5.0
<i>Part 2 – Scientific correctness</i>	
Predicted ozone yield values look realistic	4.5
Reported model error is good enough for practical use	4.7
<i>Part 3 – Overall</i>	
Would trust the tool to suggest starting points for new experiments	5.0
Dashboard is overall a useful tool for KapiteinLabs	5.0

All KapiteinLabs reviewers gave the maximum score on every usability and overall question, and the scientific correctness questions averaged 4.6 across the reviewers who felt confident answering them. These ratings should be read with the important context that all three structured-form reviewers are colleagues at KapiteinLabs, which may bias the scores towards the positive end, so the open comments and the external feedback are at least as informative as the numerical ratings.

What worked well. The interface was described as clear and well-organised, with the input ranges and units shown directly under each field highlighted as a clarity feature. The Inverse Optimisation tab received the strongest positive feedback: one reviewer described it as making “a complex machine learning task seem incredibly simple to perform”, and another pointed to it as a particular strength of the dashboard. The model dropdown labels with plain-English tags (*fast prediction*, *advanced fast prediction*, *complex prediction*, *physics-powered prediction*) were noted as helpful for choosing a model without needing to know the technical machine learning details. The

predicted-versus-actual scatter plot in the *About this model* panel was also called out as a useful way to see the model's accuracy.

What could be improved. The most consistent piece of constructive feedback was that MAE and MAPE are reported without an explanation of what they mean for users without a machine learning background. This feedback led to a direct change in the dashboard: short plain-English explanations of MAE and MAPE were added to the *About this model* panel, as described in Section 5.2.

External perspective from a plasma reactor researcher. The dashboard was also shown to a researcher who designs and operates pulsed plasma reactors as part of their daily work, including running the experiments that produced the Campaign 5 dataset. The first impression was positive on both the design and the overall idea. Forward prediction was highlighted as the more useful tab for that researcher's own work, the data upload feature was supported as a good addition, and the level of detail in the results was considered appropriate. One observation was that there are physical thresholds below which plasma ignition cannot occur, which is consistent with the operational risks discussed in Section 4.4.

6 — Conclusion and Recommendations

6.1 Conclusions

The goal of this project was to develop a machine learning pipeline that supports the design of plasma reactors at KapiteinLabs by predicting ozone output from reactor settings and recommending settings for a target ozone output. The three research questions posed in Section 1.4 are all answered. The four non-linear models trained on the Campaign 5 dataset reach $R^2 \geq 0.968$ on the test set, with the PINN achieving the lowest error (MAE = 1.48 g/kWh, MAPE = 1.9%), so a machine learning model can predict ozone yield from the controllable reactor parameters with high accuracy (RQ1). Adding monotonicity constraints to the neural network produced a small but consistent improvement on every central tendency metric and aligned the model with known physics, which confirms that physical knowledge is useful as a regularisation signal even when the data-driven model is already accurate (RQ2). The inverse optimisation framework, built on top of the trained models, hits user-defined yield targets to within 0.01 g/kWh in most cases, which is well below the model’s own MAE and confirms that the tool can recommend reactor settings for a target ozone output within the trained parameter range (RQ3). The full deliverable, comprising the data pipeline, the trained models, the inverse optimisation framework, and the Streamlit dashboard, gives KapiteinLabs a working tool that can be used by both the technical team and external clients without writing any code.

6.2 Recommendations

Two recommendations follow from the work described in this report.

The first is to **expand the dataset**. The current model is trained on six experimental sessions covering only two reactor diameters (40 and 50 mm), two flow rates (10 and 20 L/min), and dry synthetic air. To make the prediction tool reliable across a wider range of designs, future experiments should fill these gaps: more reactor diameters, more flow rates, humid air, and ideally repeat measurements at the same settings to estimate the noise floor of the experimental setup. The dashboard already supports retraining on new data through the Data Management section, so the model can grow as new measurements arrive without any code changes.

The second recommendation is to **apply the same approach to other gas treatment applications**. The pipeline developed in this project, raw data processing, model training with physics-informed regularisation, and inverse optimisation through a dashboard, is not specific to ozone production. The same approach could be used to model and optimise plasma-based treatment of other gas streams, such as NO_x removal, volatile organic compound destruction, or CO₂ conversion. The underlying machine learning code transfers directly. Only the dataset and the physics constraints would need to be adapted to the new application. This would let KapiteinLabs reuse the engineering investment from this project across a broader product portfolio.

6.3 Reflection

Looking back at the six months of this project, the part I am most proud of is delivering a working tool end to end. Starting from raw, unstructured oscilloscope and spectrometry files and ending with a deployed dashboard that non-technical users can operate, I handled every stage myself: data preparation, exploratory analysis, model development, hyperparameter tuning, inverse optimisation, and deployment. Owning the full pipeline gave me a clear view of how the pieces connect and how decisions early in the data work shape what is possible later in the modelling.

The hardest part was the data preprocessing. The raw electrical and spectrometry files were highly unstructured, with no consistent documentation, per-date exceptions in the sweep order, and a MATLAB analysis pipeline that had to be reverse-engineered before I could process anything. This is work that would normally sit closer to electrical engineering than to data science, and I had to learn enough about the measurement setup to make sure the Python pipeline reproduced the original analysis correctly. It took longer than I expected, but seeing the verification plots match the MATLAB reference was one of the most satisfying moments of the project.

Working on a tool that real engineers will use to support real design decisions also

gave me a clearer sense of the ethical side of applied machine learning. The model produces a single number for every set of inputs, and that number looks confident on the screen even when the inputs sit at the edge of the trained range. Designing the dashboard around that fact, by showing MAE and MAPE next to every prediction, flagging out-of-range inputs, and framing the optimiser's recommendations as starting points rather than answers, was as much an ethical choice as a design choice. The same applies to the data: using an externally collected dataset comes with a responsibility to report its boundaries honestly and to consult the original experimenters when making decisions that affect what stays in or out. These are not things I would have thought about in the same way before this project.

Overall, the project taught me that the value of a data science deliverable comes from the full chain, not just the model. A high-accuracy model is not useful if the data behind it is wrong, and a correct model is not useful if the people who need it cannot interact with it. Building something that engineers and clients can actually use, on top of data they can trust, is what I take from this project into whatever comes next.

Bibliography

- Batakliiev, T., Georgiev, V., Anachkov, M., Rakovsky, S., and Zaikov, G. E. (2014). Ozone decomposition. *Interdisciplinary Toxicology*, 7(2):47–59.
- Chen, T. and Guestrin, C. (2016). XGBoost: A scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 785–794, San Francisco, CA, USA. ACM.
- Das, S. and Suganthan, P. N. (2011). Differential evolution: A survey of the state-of-the-art. *IEEE Transactions on Evolutionary Computation*, 15(1):4–31.
- Friedman, J. H. (2001). Greedy function approximation: A gradient boosting machine. *The Annals of Statistics*, 29(5):1189–1232.
- Grinsztajn, L., Oyallon, E., and Varoquaux, G. (2022). Why do tree-based models still outperform deep learning on typical tabular data? In *Advances in Neural Information Processing Systems 35 (NeurIPS 2022), Datasets and Benchmarks Track*, pages 507–520.
- Huiskamp, T., Pemen, A. J. M., and van Heesch, E. J. M. (2016). Matching a (sub)nanosecond pulse source to a corona plasma reactor. *IEEE Transactions on Plasma Science*, 44(10):2240–2248.
- Huiskamp, T., Pemen, A. J. M., and van Heesch, E. J. M. (2017). Subnanosecond transient plasma ozone generation and NO removal. *IEEE Transactions on Plasma Science*, 45(4):610–619.
- Jiang, B., Zheng, J., Qiu, S., Wu, M., Zhang, Q., Yan, Z., and Xue, Q. (2014). Review on electrical discharge plasma technology for wastewater remediation. *Chemical Engineering Journal*, 236:348–368.
- Kebriaei, M., Ketabi, A., and Halvaei Niasar, A. (2015). Pulsed corona discharge, a new and effective technique for water and air treatment. *Biological Forum – An International Journal*, 7(1):1686–1692.
- Li, J., Xu, J., Rebrov, E. V., and Bogaerts, A. (2025). Machine learning-based prediction and optimization of plasma-catalytic dry reforming of methane in a dielectric barrier discharge reactor. *Chemical Engineering Journal*, 507:159897.
- Raissi, M., Perdikaris, P., and Karniadakis, G. E. (2019). Physics-informed neural networks: A deep learning framework for solving forward and inverse problems in-

- volving nonlinear partial differential equations. *Journal of Computational Physics*, 378:686–707.
- Shi, H., Wang, S., and Wang, P. (2024). From simulation to reality: CFD-ML-driven structural optimization and experimental analysis of thermal plasma reactors. *Journal of Environmental Chemical Engineering*, 12(3):112998.
- Storn, R. and Price, K. (1997). Differential evolution – a simple and efficient heuristic for global optimization over continuous spaces. *Journal of Global Optimization*, 11(4):341–359.
- Ton, M. (2021). The influence of high-voltage pulse parameters and reactor dimensions on transient plasma ozone production in cylinder-wire reactors. Master’s thesis, Eindhoven University of Technology.
- Zhong, L., Wu, B., and Wang, Y. (2022). Low-temperature plasma simulation based on physics-informed neural networks: Frameworks and preliminary applications. *Physics of Fluids*, 34(8):087116.

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During the preparation of this thesis, I used the following generative AI tool(s): Claude (Anthropic).

I used these tool(s) for the following purposes: improving the clarity and grammar of text I had already written myself; reviewing, debugging, and improving the readability of code I had already written; summarising papers I had already read to support my own understanding; and giving me feedback on ideas and interpretations I had already formed myself.

I did not use generative AI to generate arguments, results, or conclusions presented as my own; to fabricate, alter, or invent data; or to produce the core analysis of this thesis.

All research design, implementation, analysis, and conclusions are my own. I have critically reviewed and edited all content produced or supported by generative AI tools. I take full responsibility for the entire content and integrity of this thesis, including any parts that were drafted, refined, or supported with the assistance of AI.

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